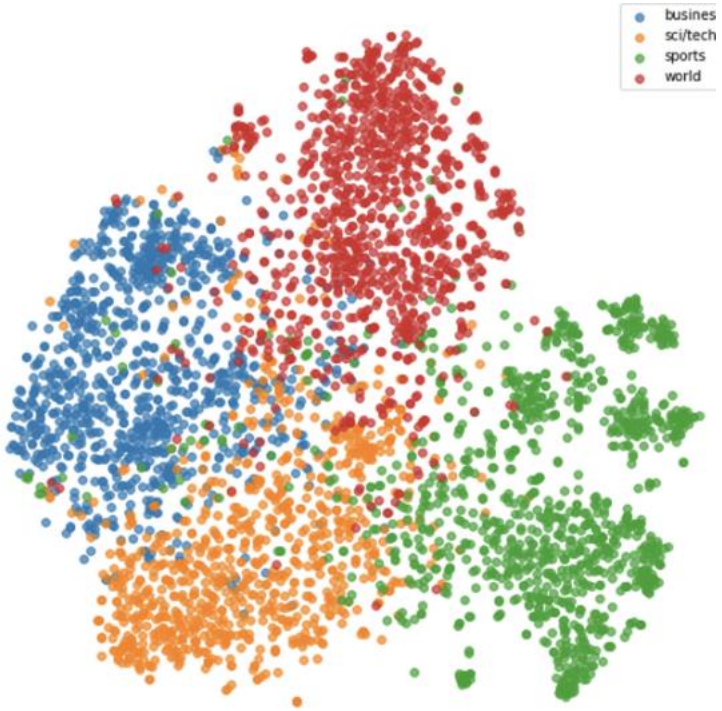
Reconocimiento de Patrones

Versión 2026



Clustering y Bag of Words

(CAPÍTULO 4 – PR.04.D)



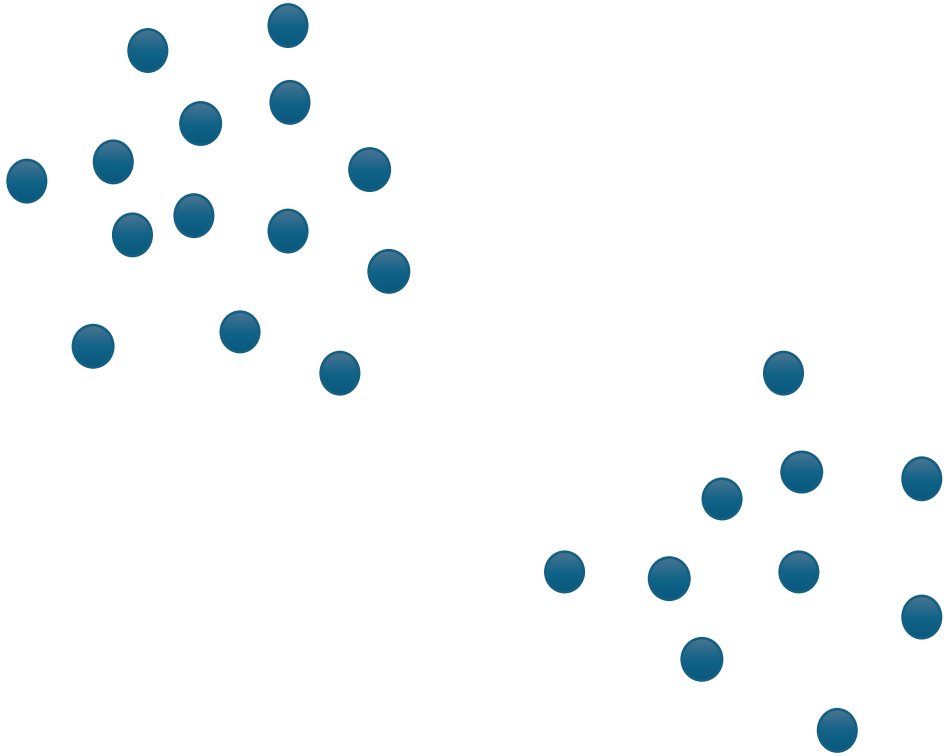
Domingo Mery

Ciencia de la Computación
Universidad Católica – Chile
http://domingomery.ing.uc.cl

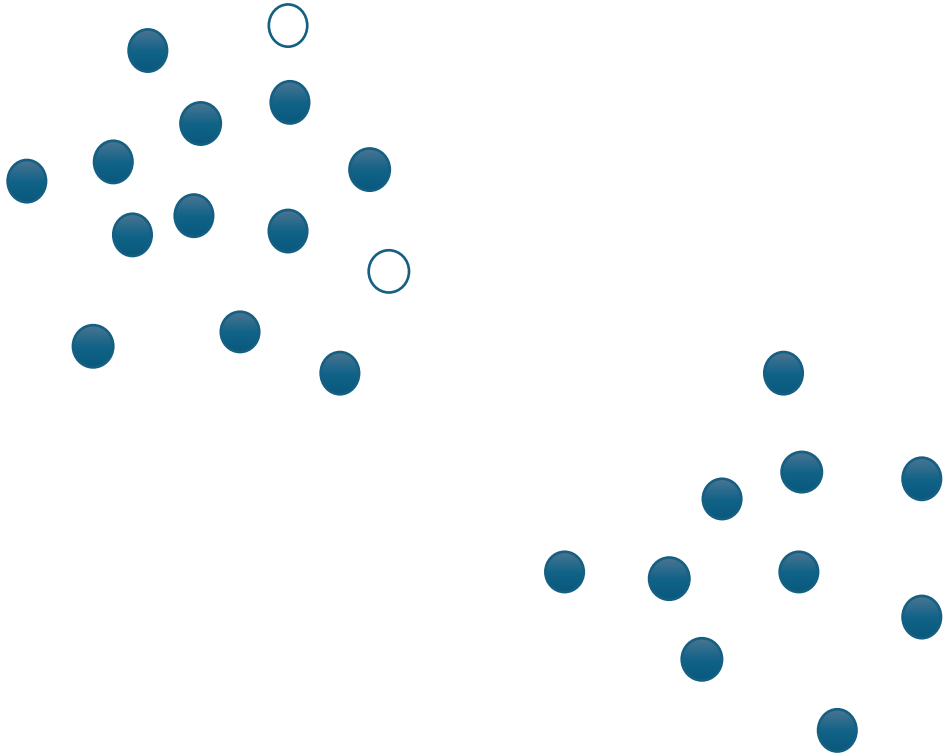
Clustering usando k-Means

Algorithm:

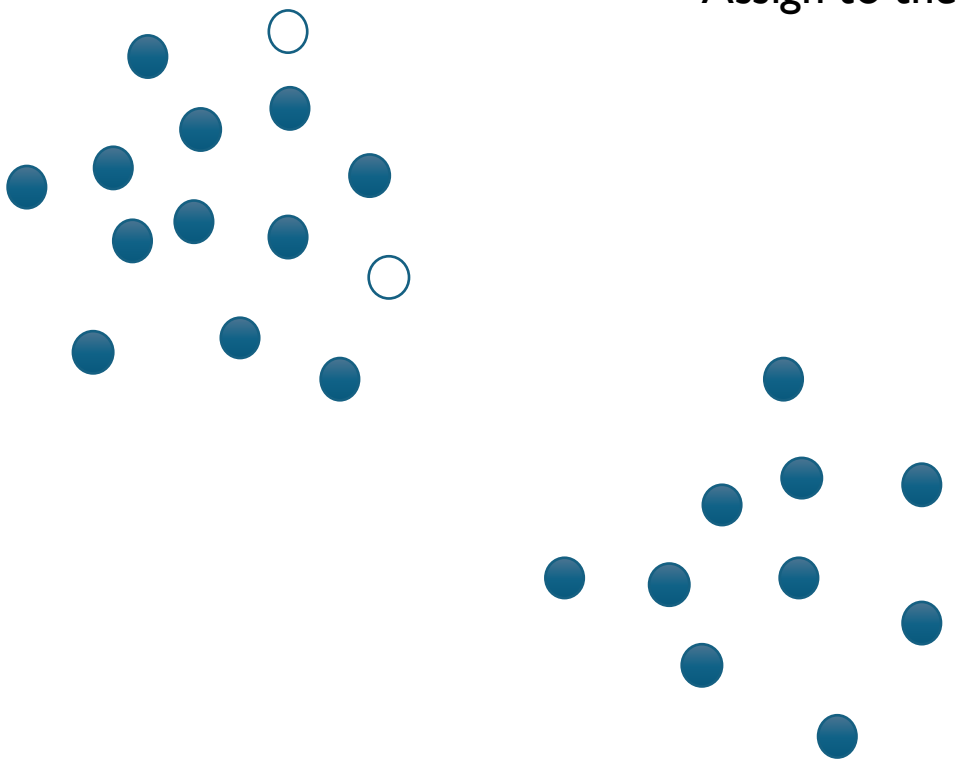
1. Input Data $X = \{x_1, x_2, \ldots x_N\}$ and number of clusters K
2. Centroids $\{c_1, c_2, \ldots c_K\}$ = random K points of X
3. For each data point x_i
4. Compute distance $d_{ij} = d(x_i,c_j)$ $i=1,\ldots,N$, $j=1,\ldots K$
5. Assign x_i to the nearest centroid: $y_i = \text{argmin}_j\{d_{ij}\}$
6. Compute the new centroids of each cluster
   $c^*_j = \text{mean}(x_i)$ for $y_i = j$
7. if $c^*_j \neq c_j$ then $c_j = c^*_j$ go to step 3
8. Output: $\{c^*_1, c^*_2, \ldots c^*_K\}$ and y_i for $i=1,\ldots,N$

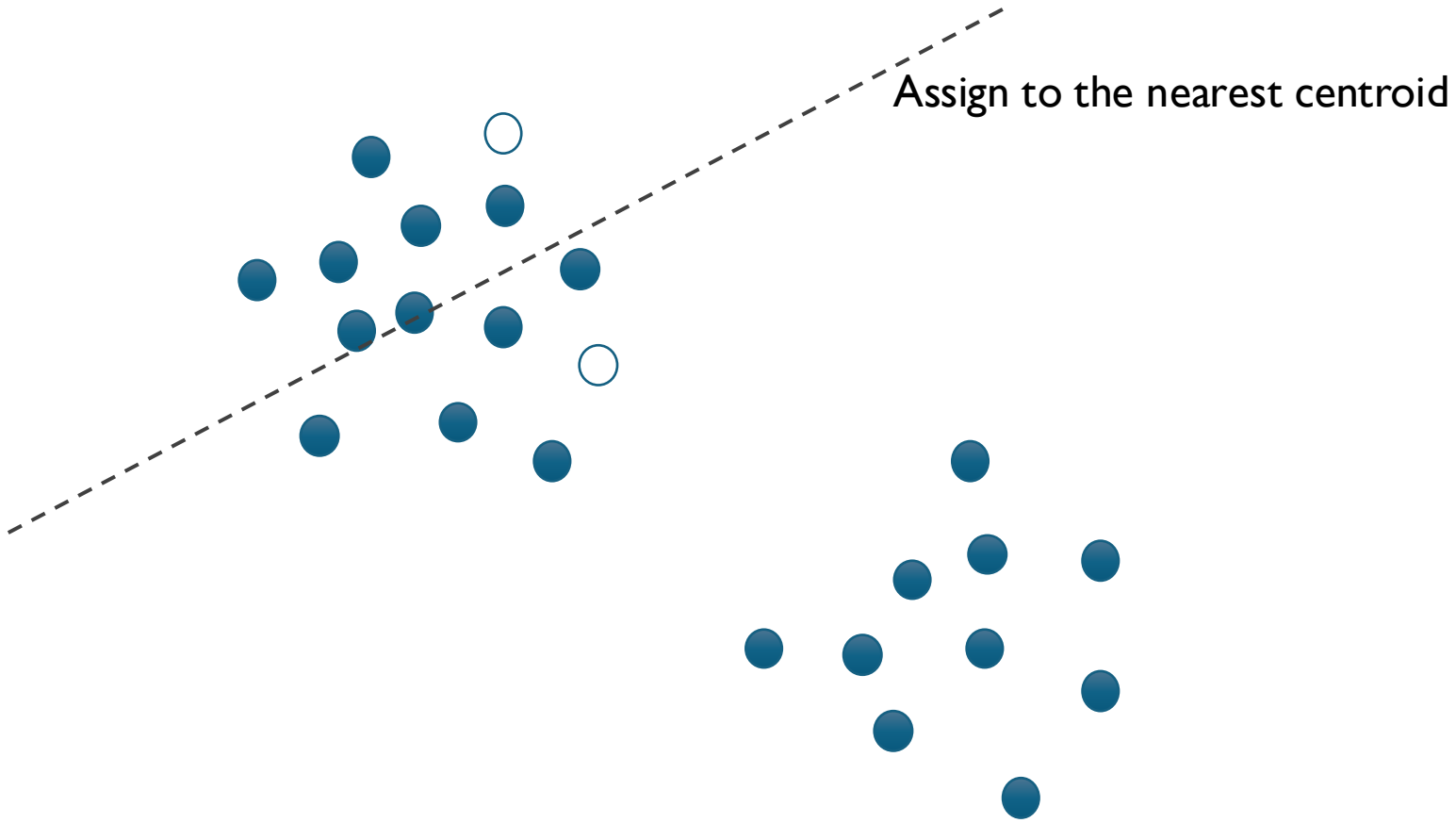


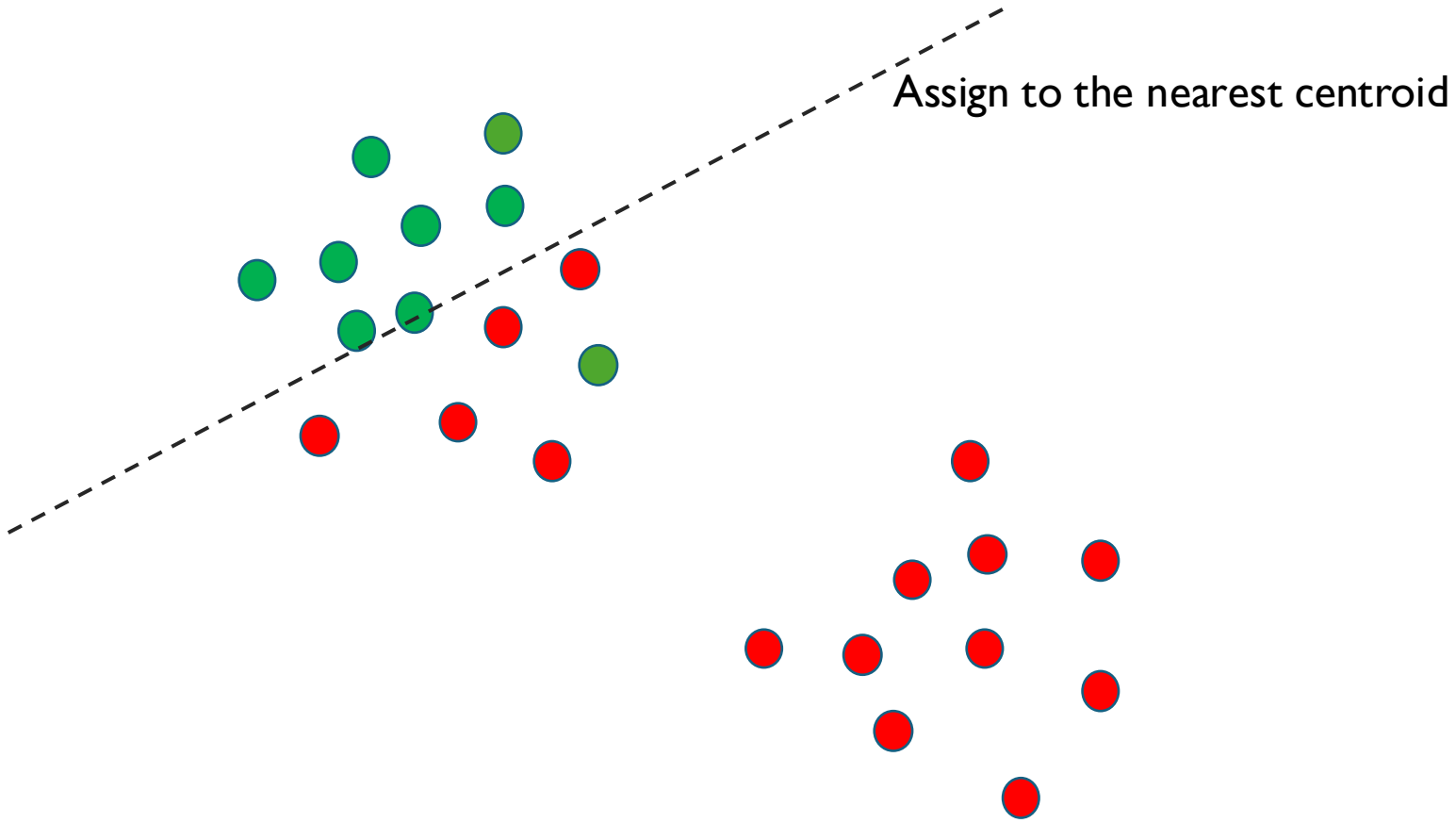
Choose random K=2 points (centroids)

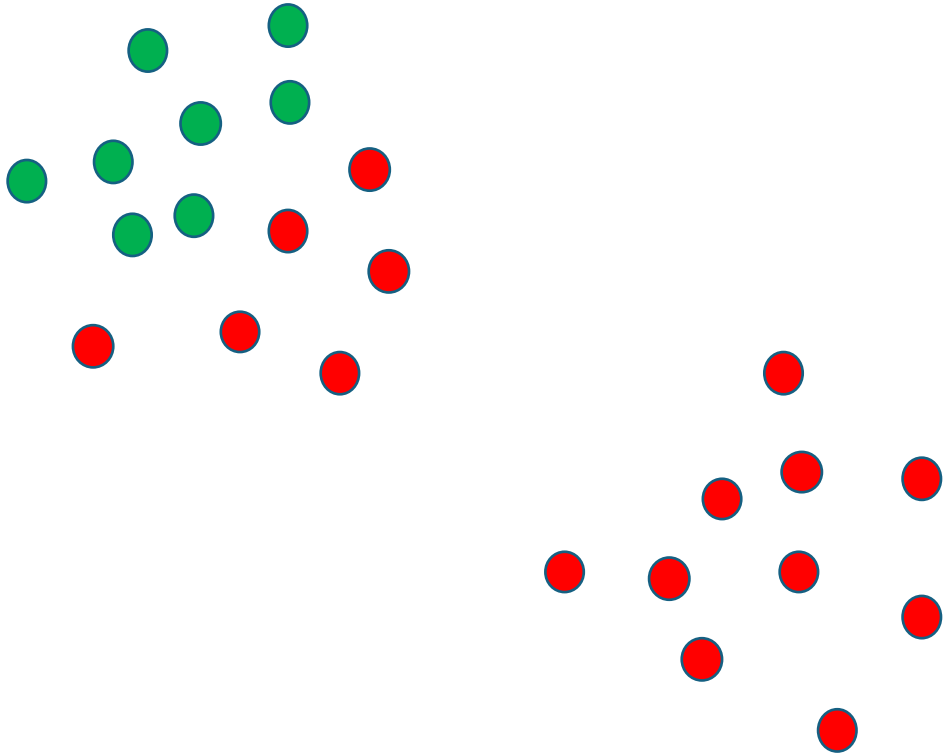


Assign to the nearest centroid

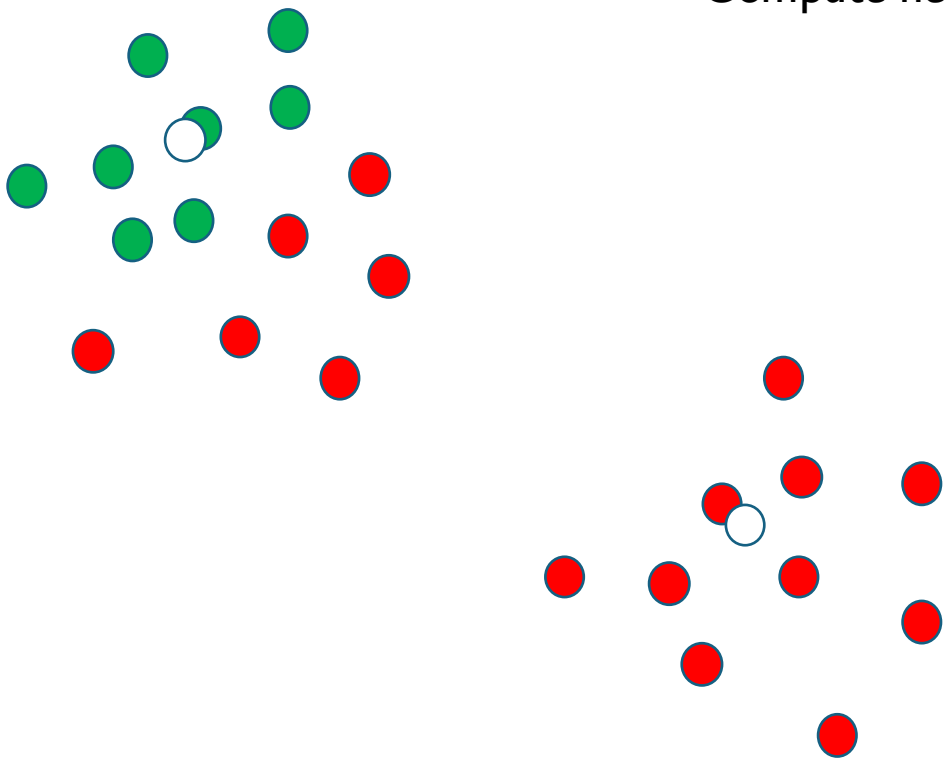


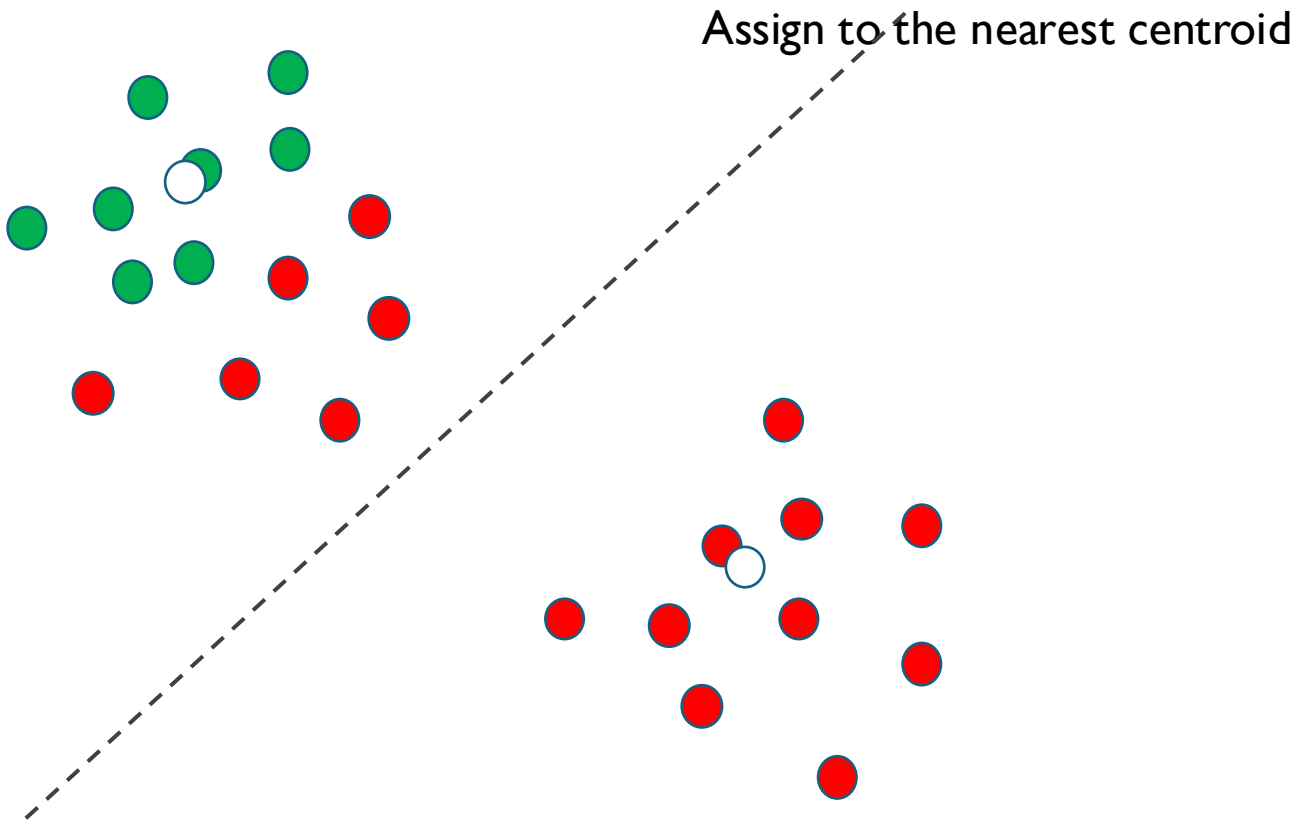

Assign to the nearest centroid

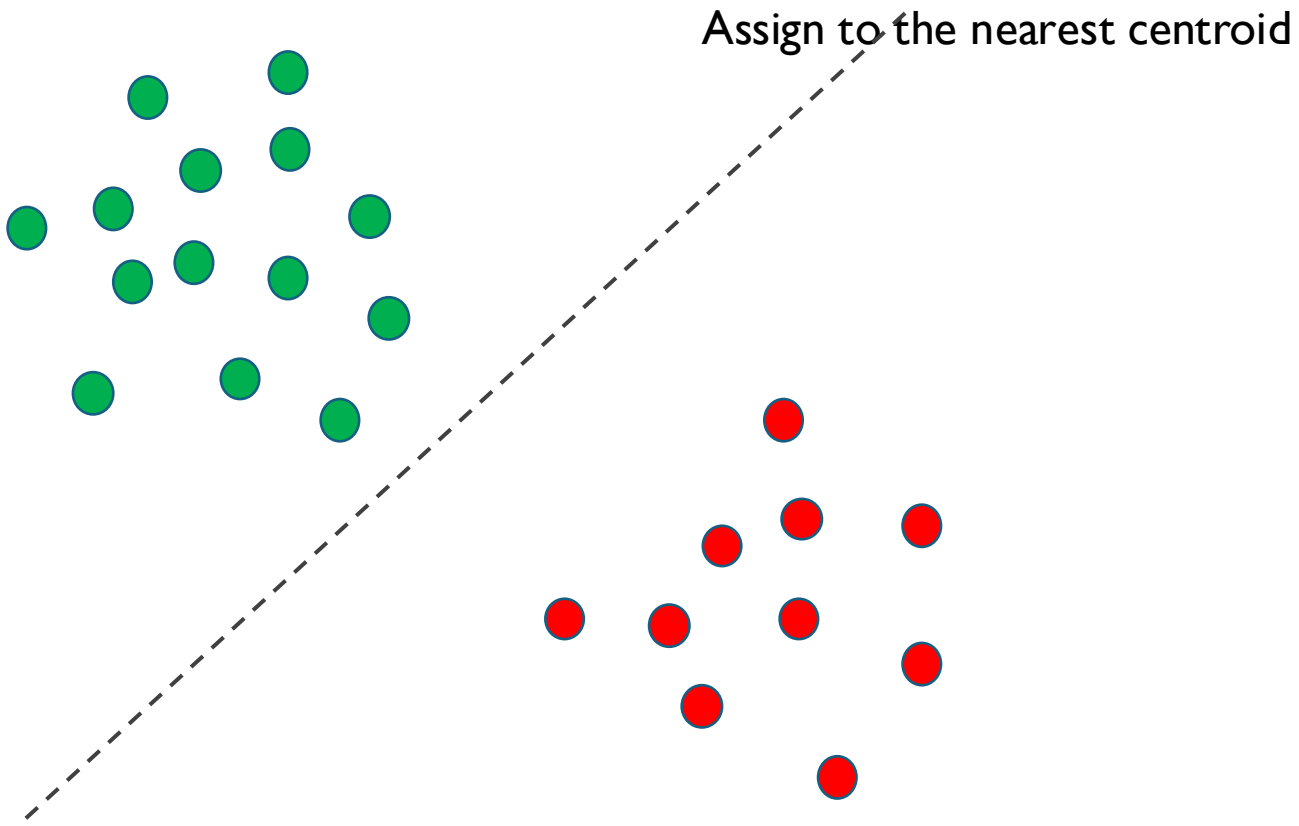

Assign to the nearest centroid



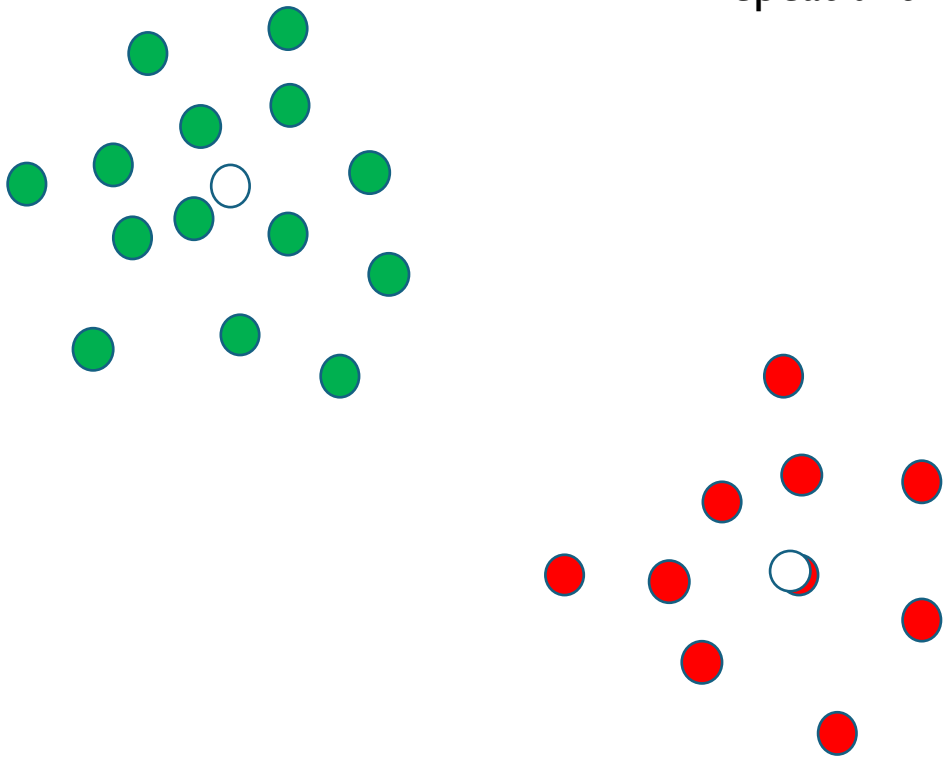
Compute new centroids




Assign to the nearest centroid


Assign to the nearest centroid

Repeat until convergence

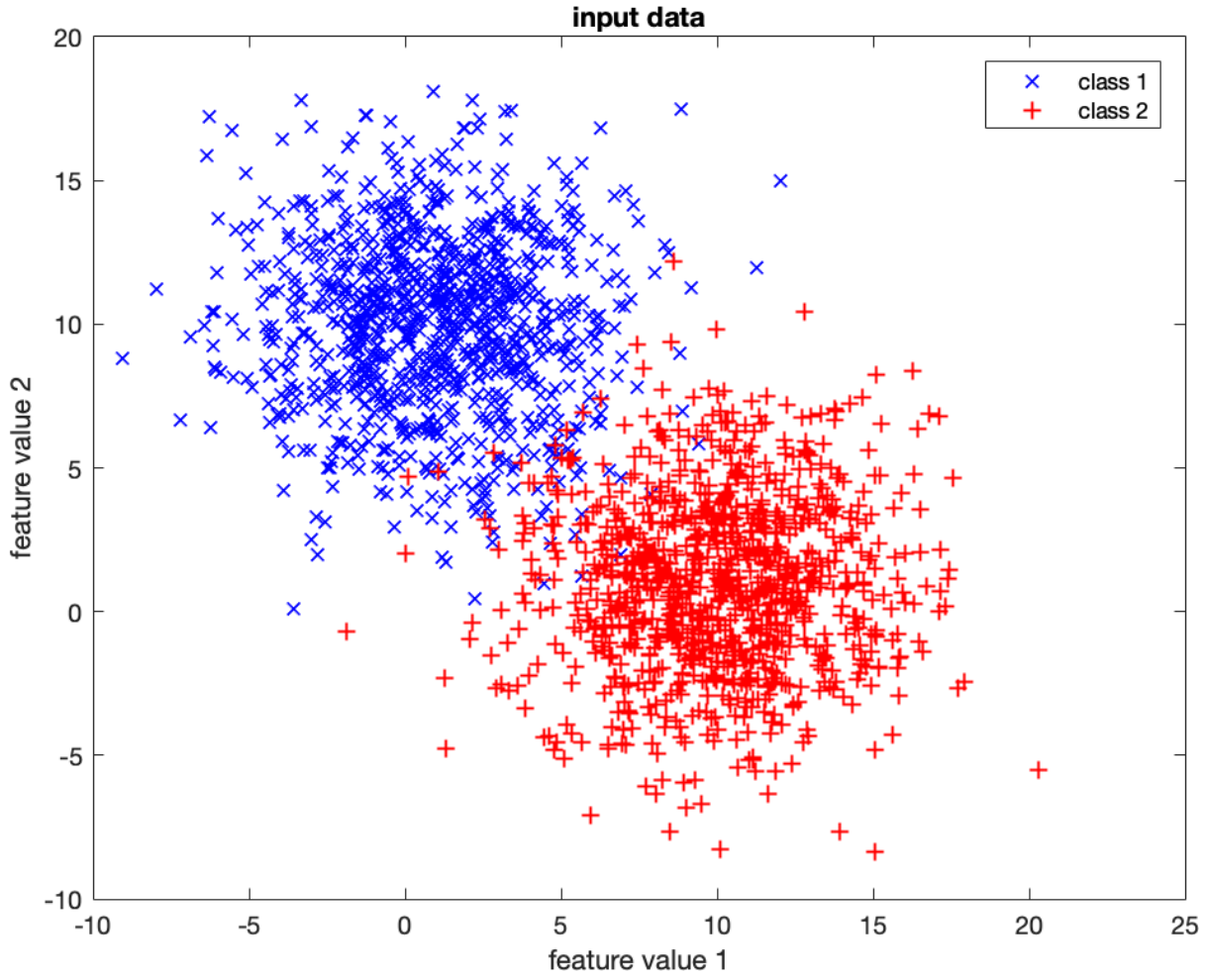


Algorithm:

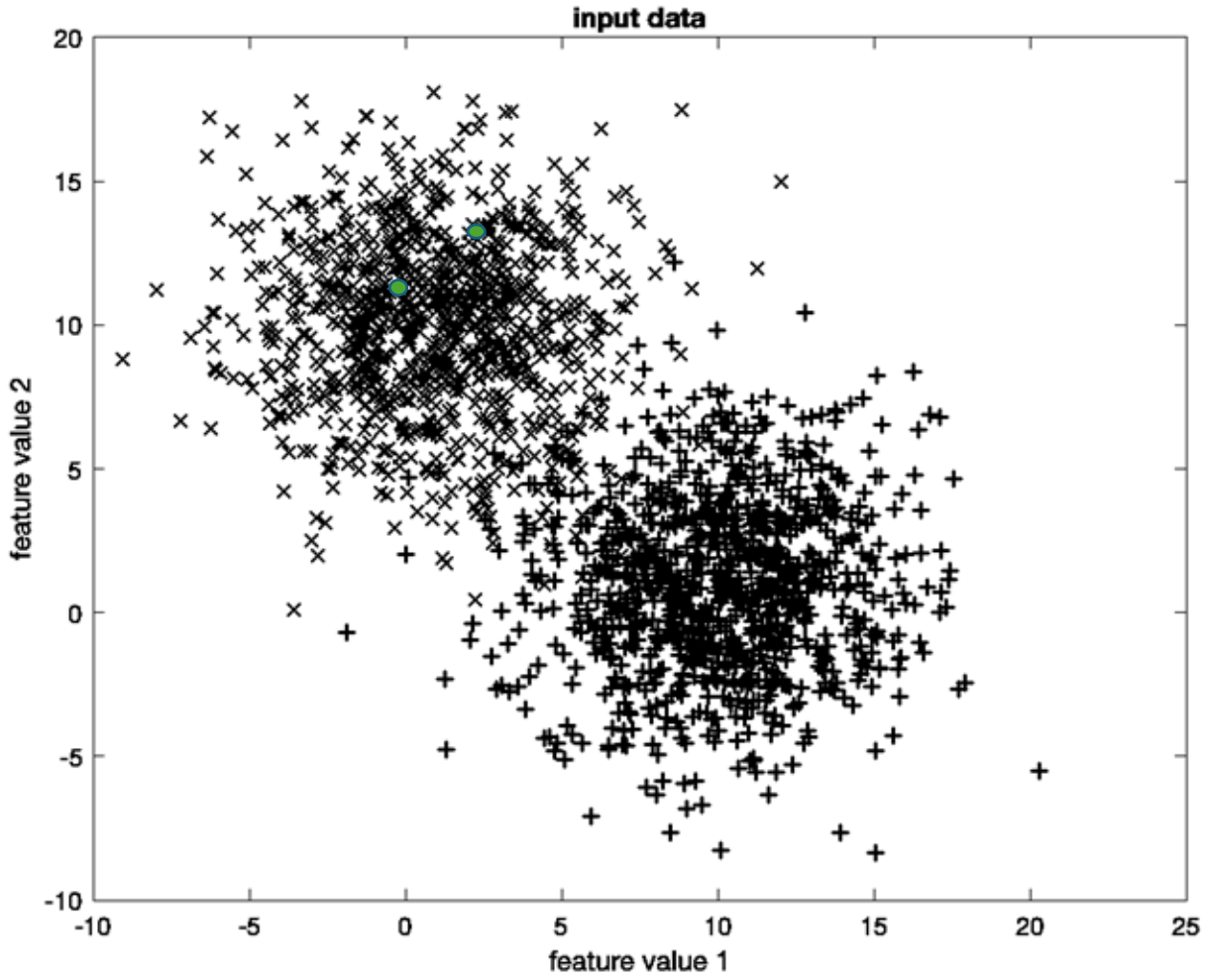
1. Input Data $X = \{x_1, x_2, \ldots x_N\}$ and number of clusters K
2. Centroids $\{c_1, c_2, \ldots c_K\}$ = random K points of X
3. For each data point x_i
4. Compute distance $d_{ij} = d(x_i,c_j)$ $i=1,\ldots,N$, $j=1,\ldots K$
5. Assign x_i to the nearest centroid: $y_i = \text{argmin}_j\{d_{ij}\}$
6. Compute the new centroids of each cluster
   $c^*_j = \text{mean}(x_i)$ for $y_i = j$
7. if $c^*_j \neq c_j$ then $c_j = c^*_j$ go to step 3
8. Output: $\{c^*_1, c^*_2, \ldots c^*_K\}$ and y_i for $i=1,\ldots,N$

Example: Two Clouds of Points

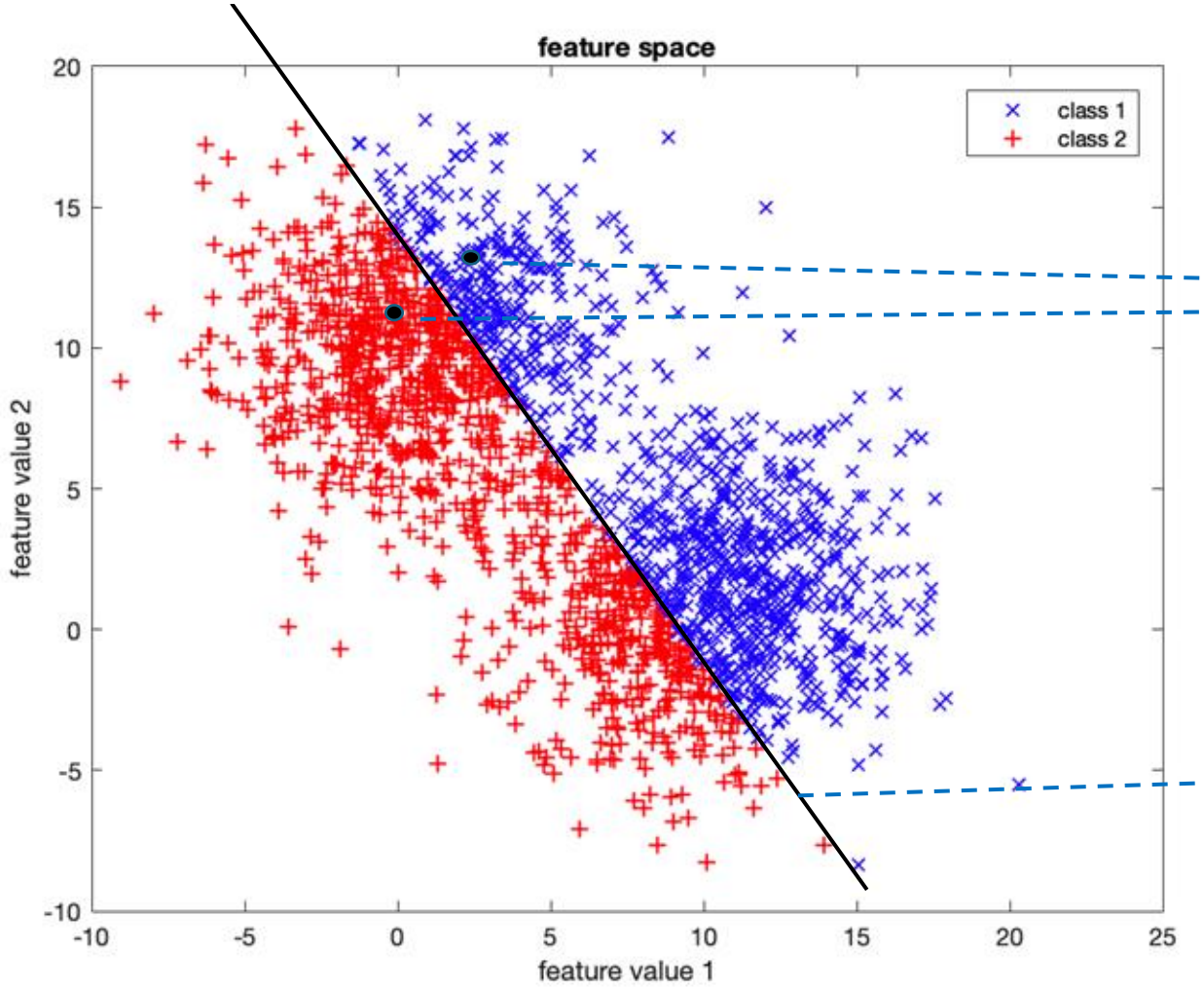
Input Data


input data
class 1
class 2
feature value 2
feature value 1
20
15
10
5
0
-5
-10
-10
-5
0
5
10
15
20
25

Iteration 0:


input data
feature value 2
feature value 1
20
15
10
5
0
-5
-10
-10
-5
0
5
10
15
20
25

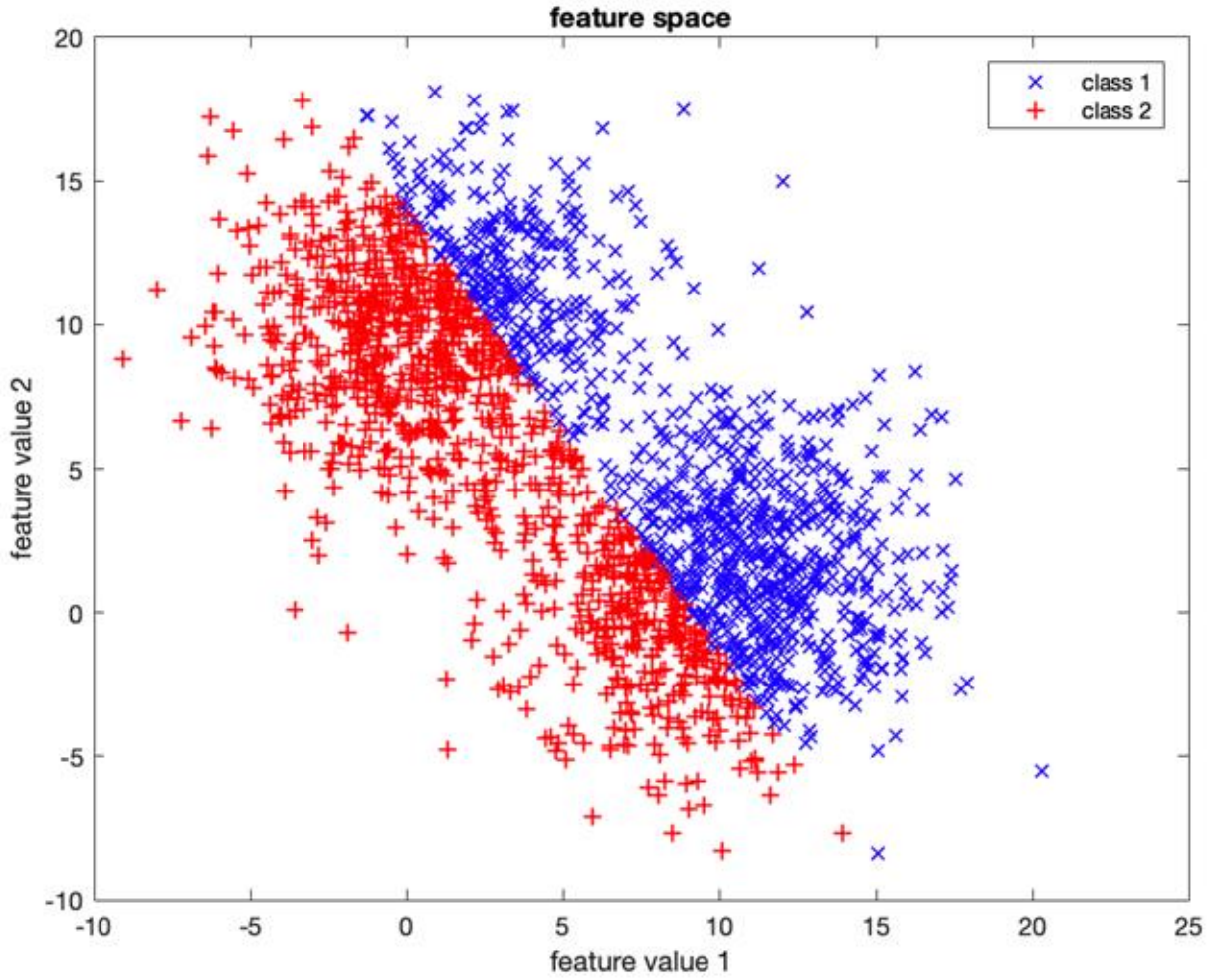
Iteration 1:



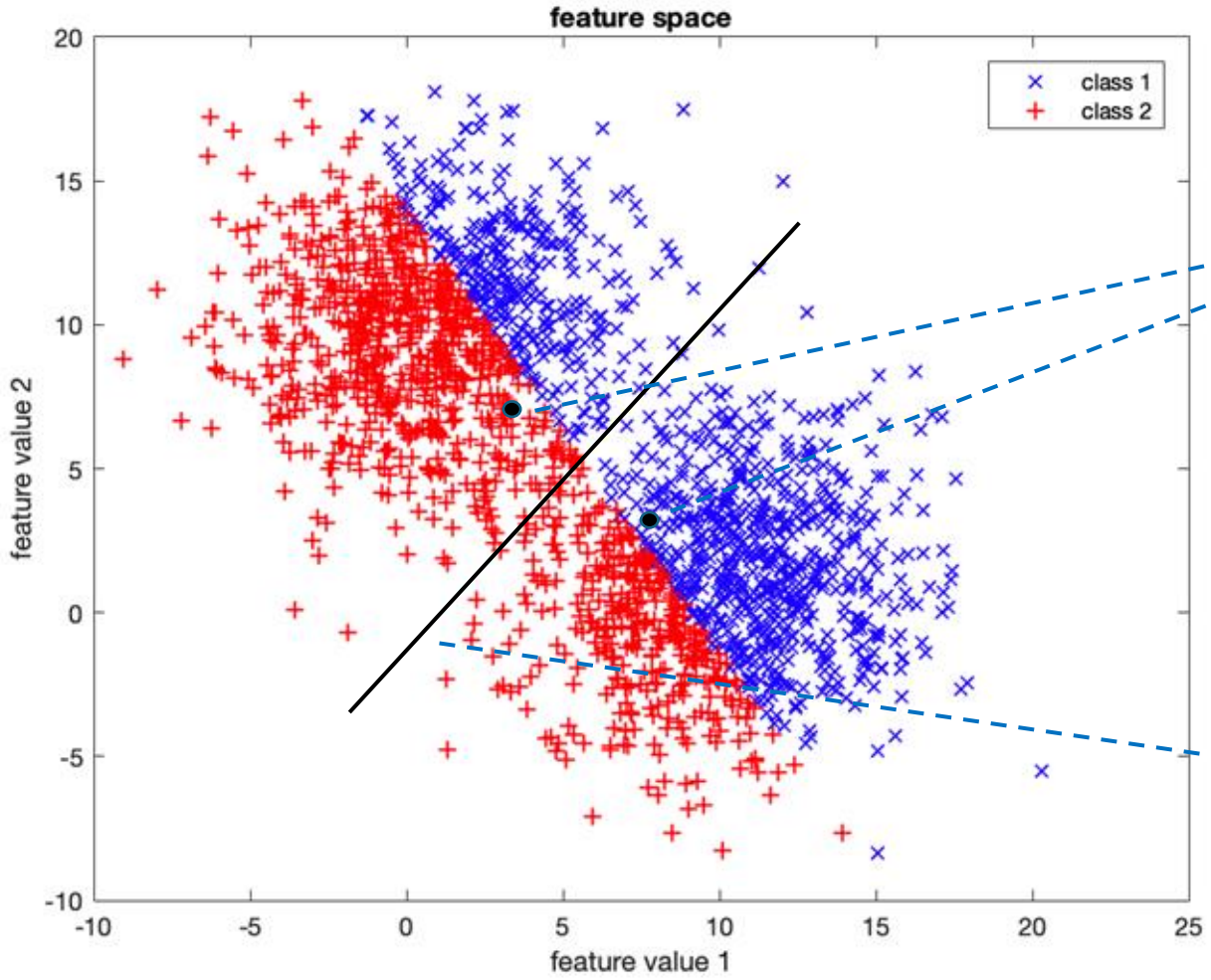
Random points

Separation line

Iteration 1:


feature space
class 1
class 2
feature value 2
feature value 1
20
15
10
5
0
-5
-10
-10
-5
0
5
10
15
20
25

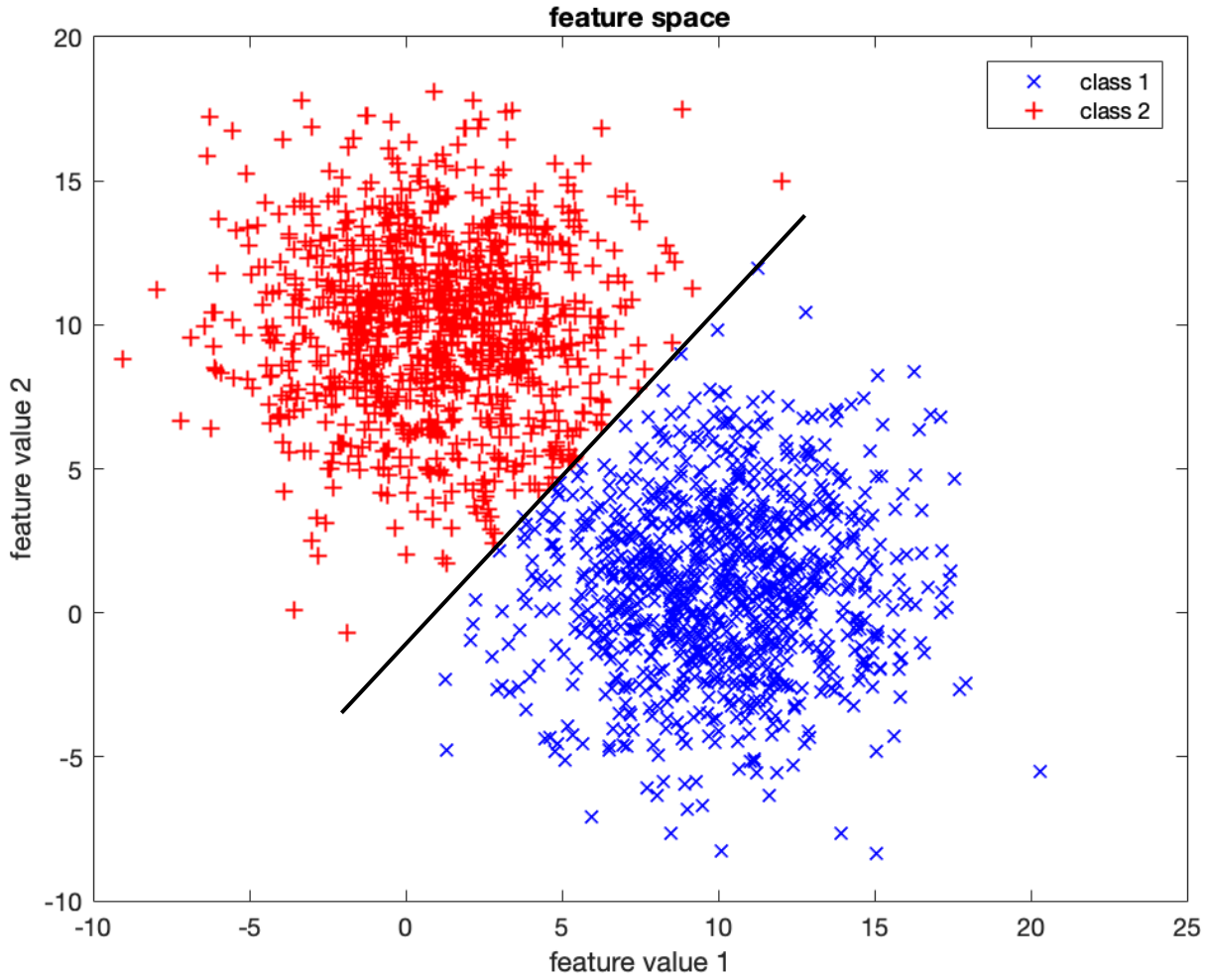
Iteration 2:



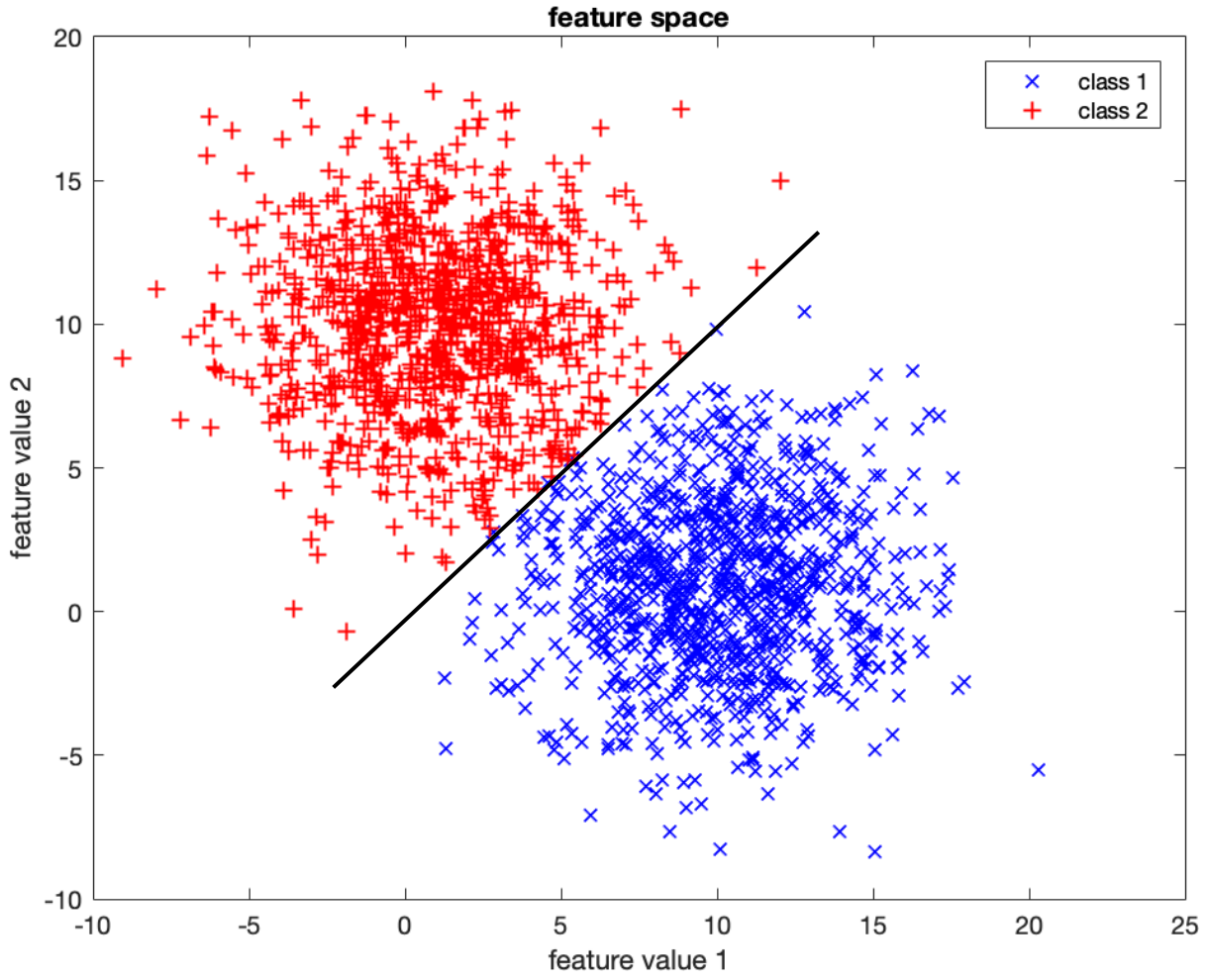
Centroids

Separation line

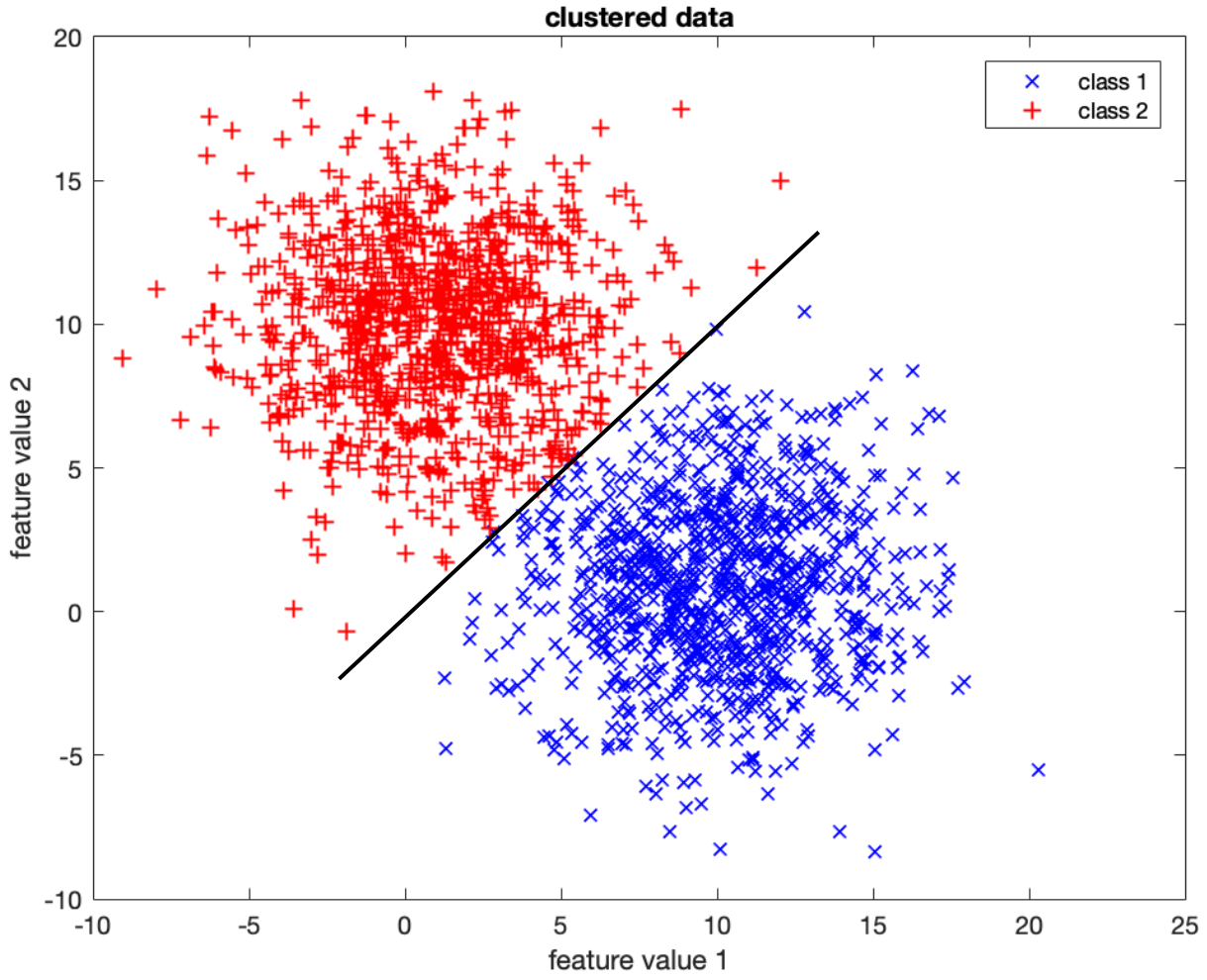
Iteration 3:


feature space
class 1
class 2
feature value 1
feature value 2
20
15
10
5
0
-5
-10
-10
-5
0
5
10
15
20
25

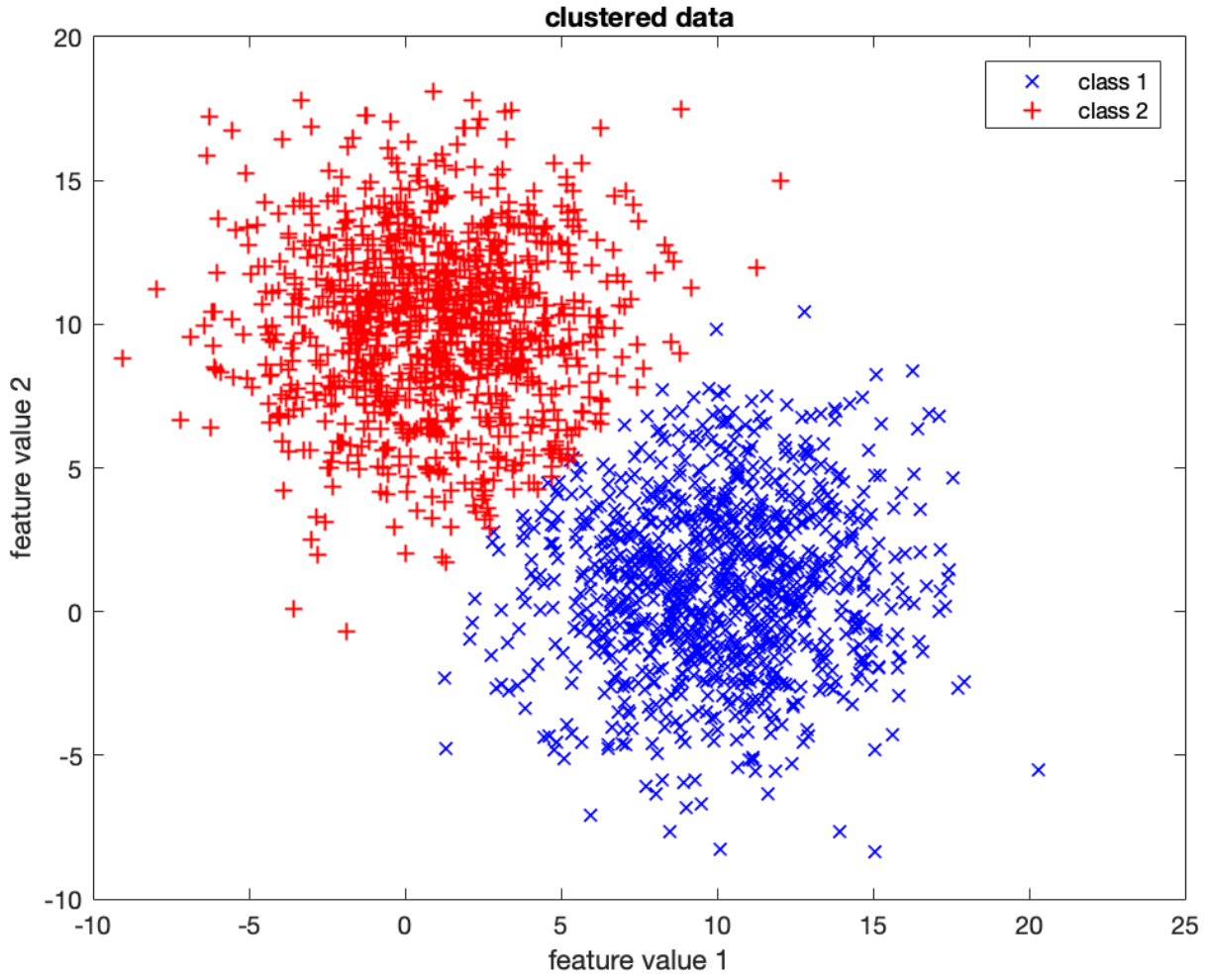
Iteration 4:


feature space
class 1
class 2
feature value 1
feature value 2
20
15
10
5
0
-5
-10
-10
-5
0
5
10
15
20
25

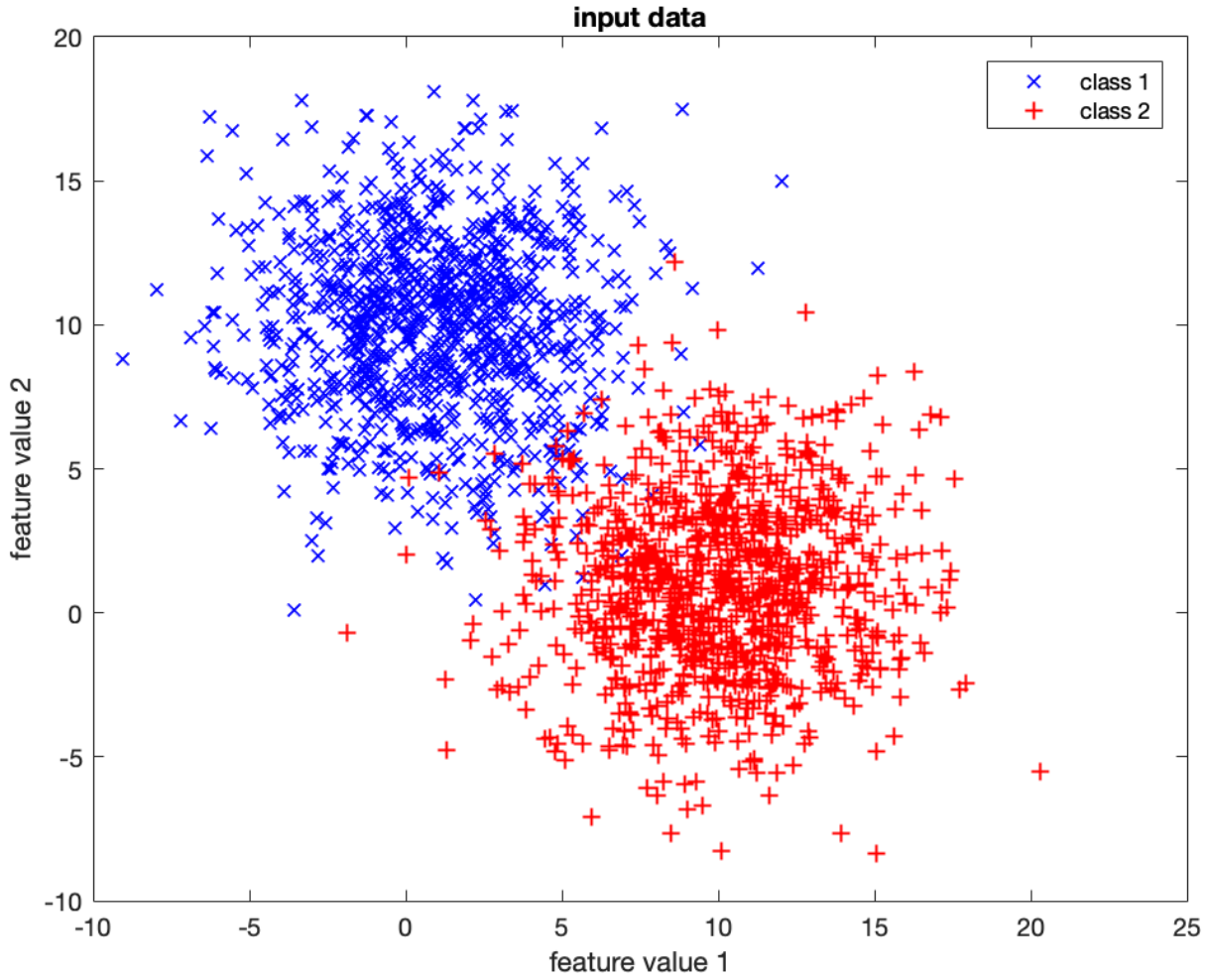
Iteration 5:


clustered data
class 1
class 2
feature value 2
feature value 1
20
15
10
5
0
-5
-10
-10
-5
0
5
10
15
20
25

Found Clusters



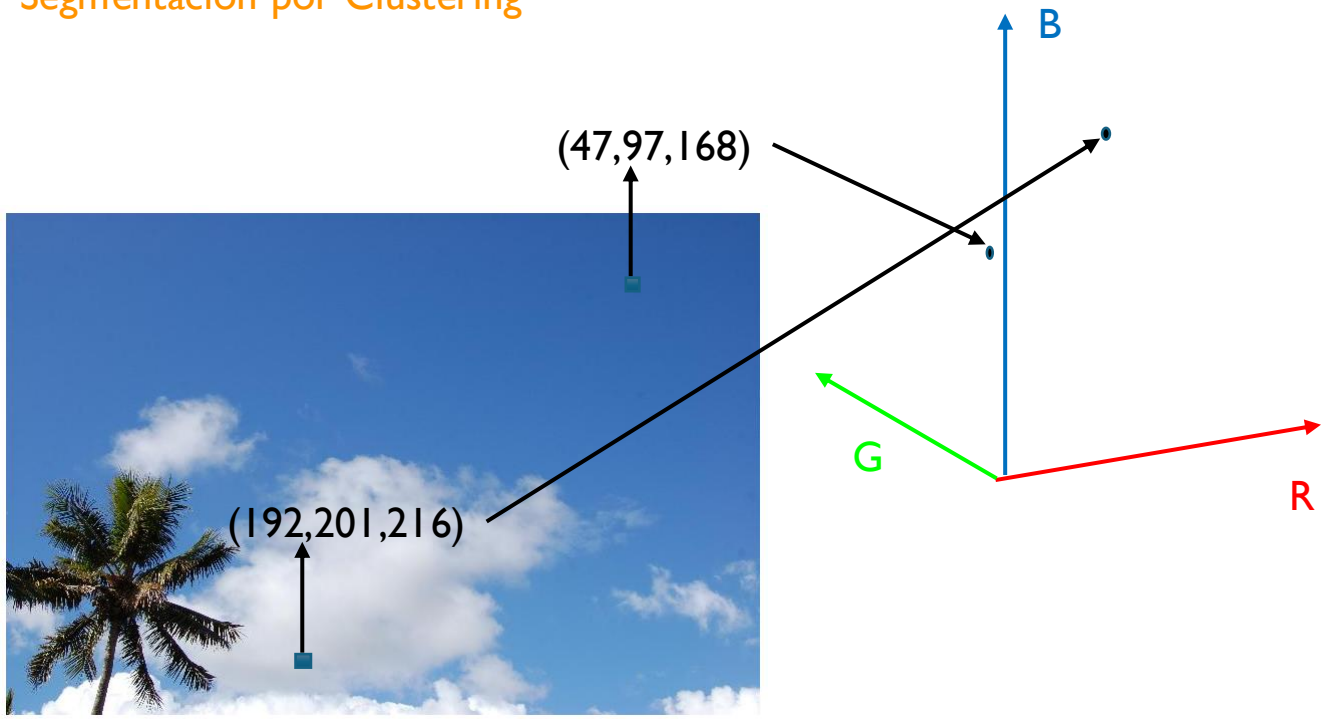
Input Data


input data
class 1
class 2
feature value 2
feature value 1
20
15
10
5
0
-5
-10
-10
-5
0
5
10
15
20
25

Example: Color Segmentation

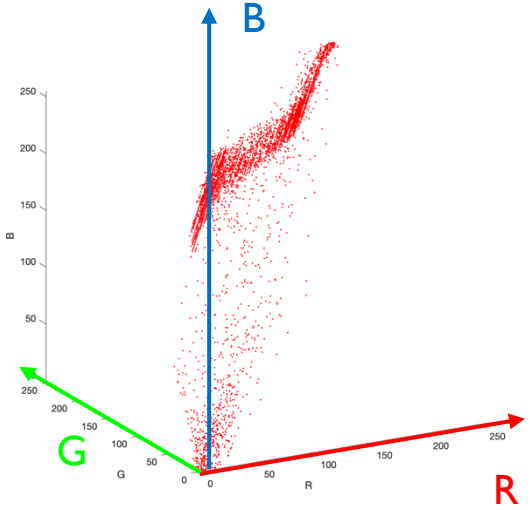


Segmentación por Clustering



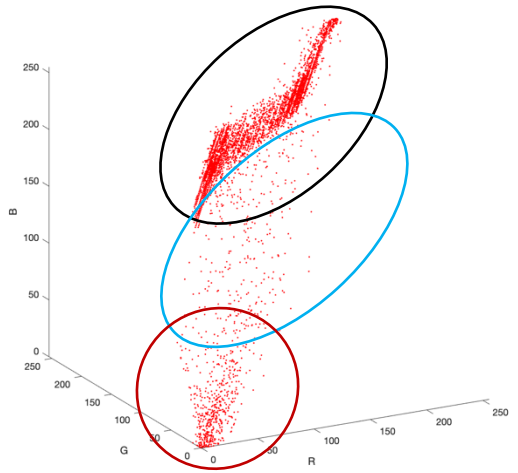
Segmentación por Clustering





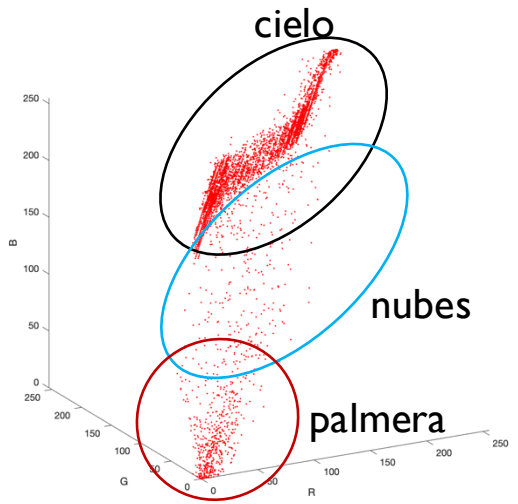
Segmentación por Clustering



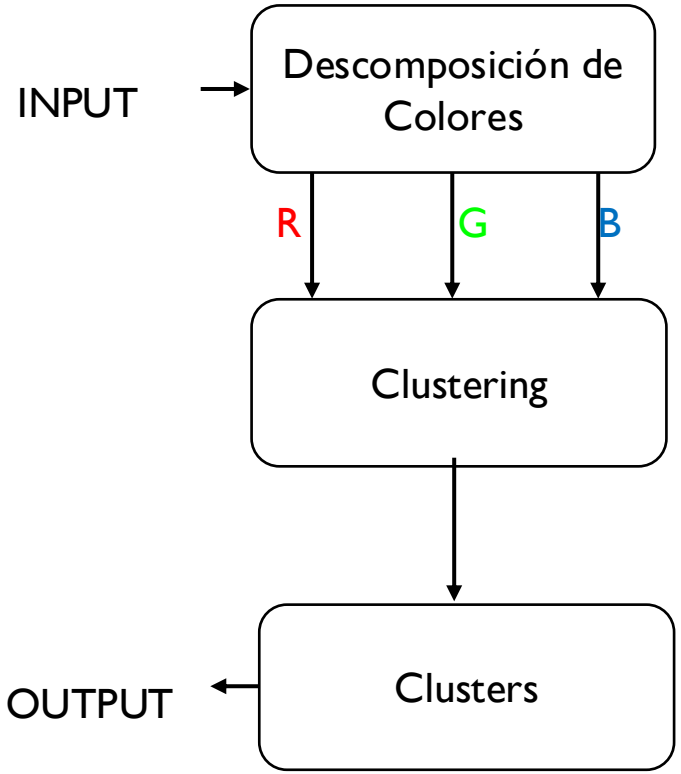


Segmentación por Clustering

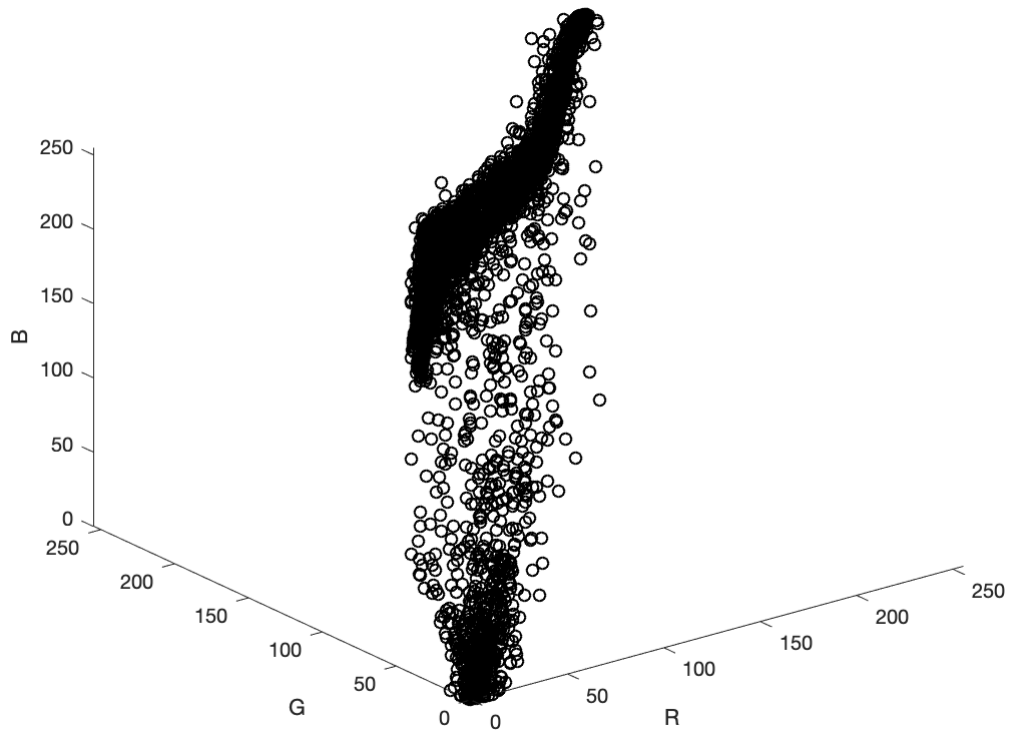


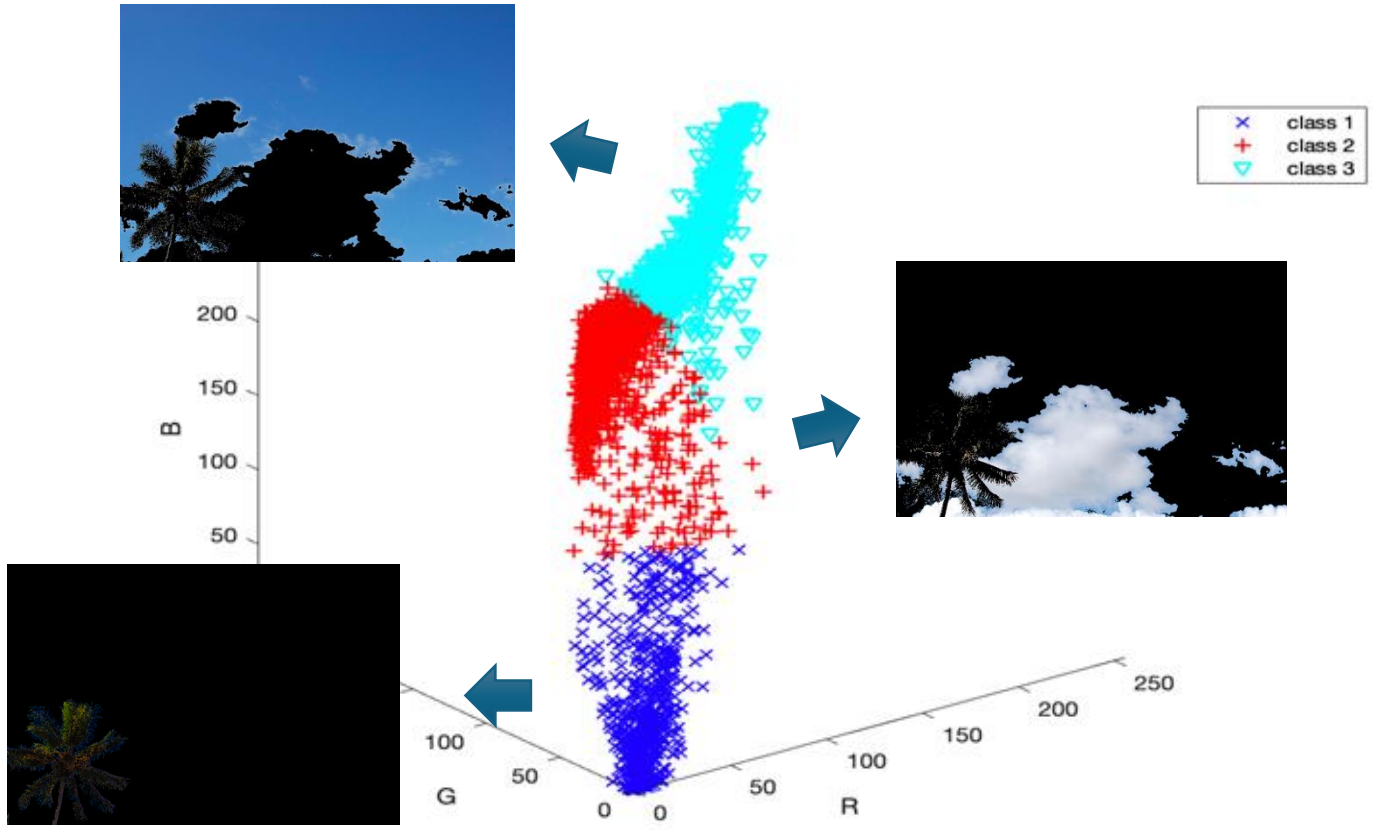


Algoritmo


INPUT
Descomposición de Colores
R
G
B
Clustering
Clusters
OUTPUT

Clustering usando k-means


B
250
200
150
100
50
0
G
250
200
150
100
50
0
R
0
50
100
150
200
250


class 1
class 2
class 3
B
200
150
100
50
G
100
50
0
0
50
100
150
200
250
R

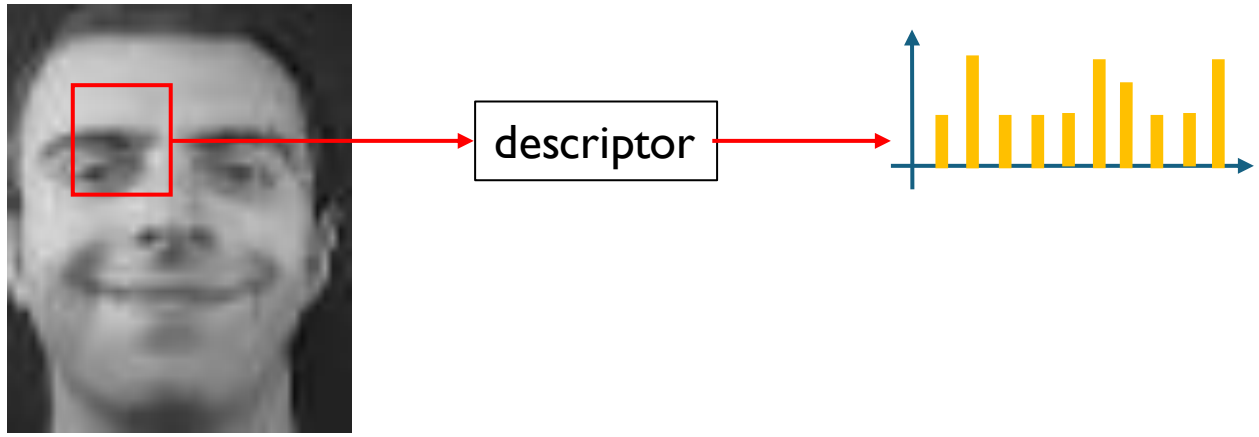

INPUT

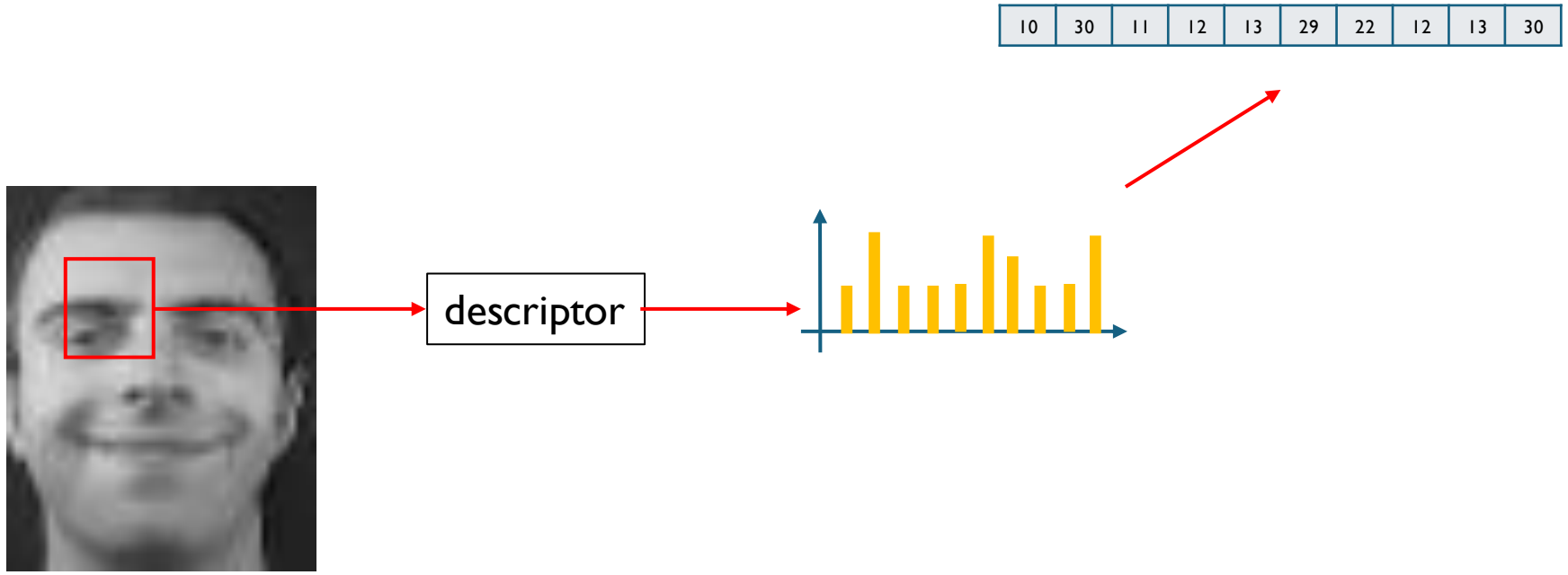

OUTPUT

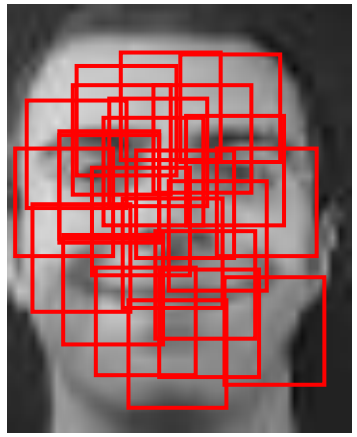
Bag of Visual Words





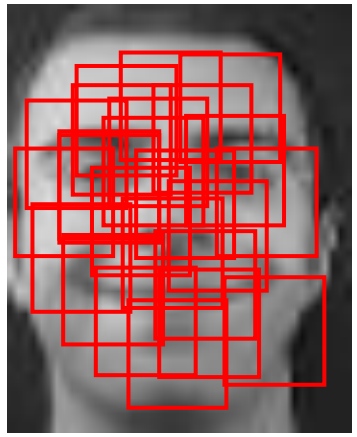

descriptor


10 30 11 12 13 29 22 12 13 30
descriptor

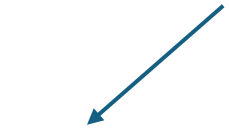


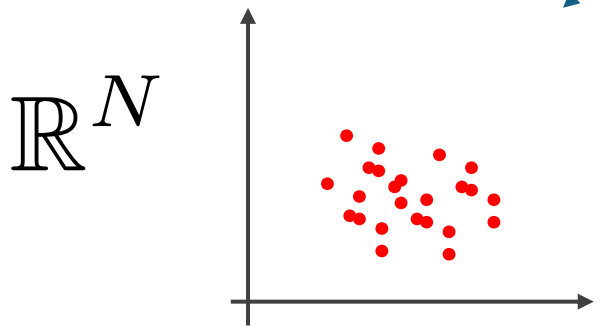
description

10	30	11	12	13	29	22	12	13	30
10	10	12	34	12	10	12	34	12	9
12	29	11	10	12	10	12	34	12	14
3	10	12	34	12	10	12	34	12	14
12	12	3	10	12	12	3	10	12	
12	10	12	34	12	10	12	34	12	14
11	10	12	34	12	10	12	34	12	14
12	29	11	10	12	10	12	34	12	14
3	10	12	34	12	12	12	3	10	12
12	12	3	10	12	12	12	3	10	12
10	10	12	34	12	10	12	34	12	14
10	10	12	34	12	10	12	34	12	9
12	29	11	10	12	10	12	34	12	14
3	10	12	34	12	10	12	34	12	14
12	12	3	10	12	12	3	10	12	15
12	10	12	34	12	10	12	34	12	14
11	10	12	34	12	10	12	34	12	14
12	29	11	10	12	10	12	34	12	14
3	10	12	34	12	12	12	3	10	12
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12	12	3	10	12	12	3	10	12	17



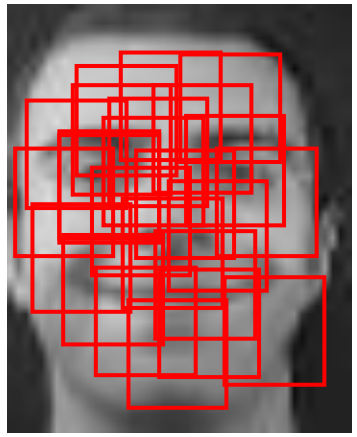
description

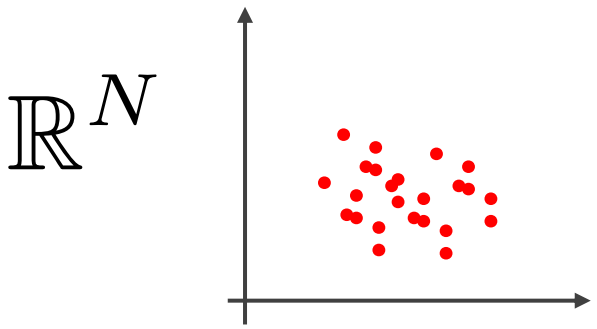


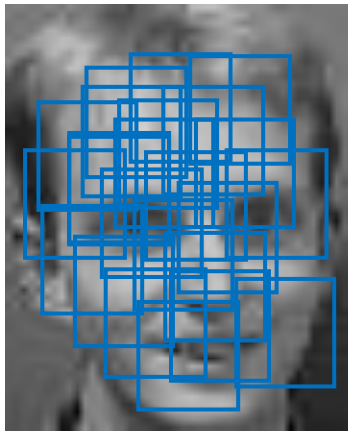


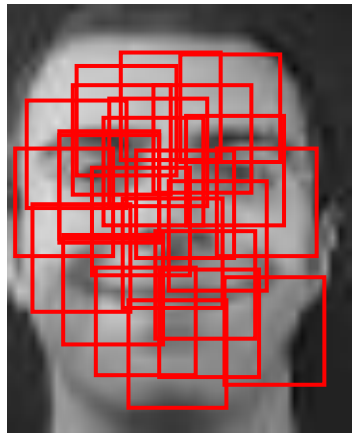
N

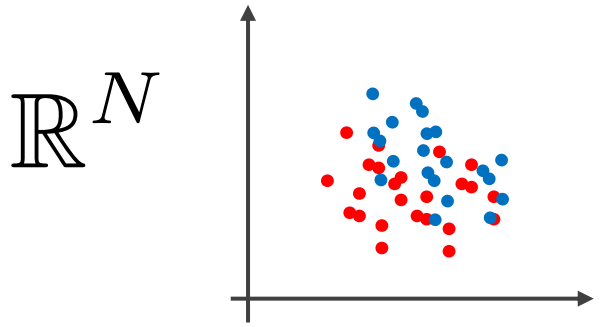
10	30	11	12	13	29	22	12	13	30
10	10	12	34	12	10	12	34	12	9
12	29	11	10	12	10	12	34	12	14
3	10	12	34	12	10	12	34	12	14
12	12	3	10	12	12	3	10	12	
12	10	12	34	12	10	12	34	12	14
11	10	12	34	12	10	12	34	12	14
12	29	11	10	12	10	12	34	12	14
3	10	12	34	12	12	12	3	10	12
12	12	3	10	12	12	12	3	10	12
10	10	12	34	12	10	12	34	12	14
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3	10	12	34	12	10	12	34	12	14
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12	10	12	34	12	10	12	34	12	14
11	10	12	34	12	10	12	34	12	14
12	29	11	10	12	10	12	34	12	14
3	10	12	34	12	12	12	3	10	12
12	12	3	10	12	12	12	3	10	12
10	10	12	34	12	10	12	34	12	14
12	12	3	10	12	12	3	10	12	17

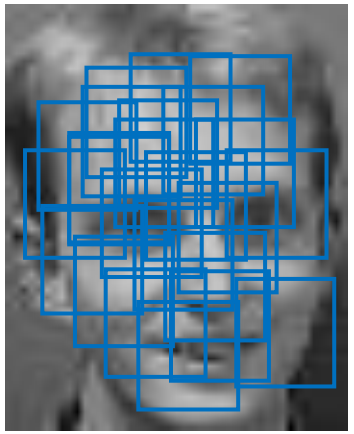


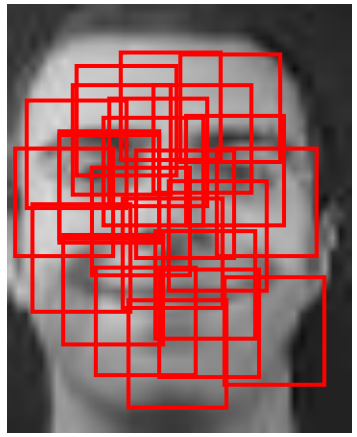

$\mathbb{R}^N$

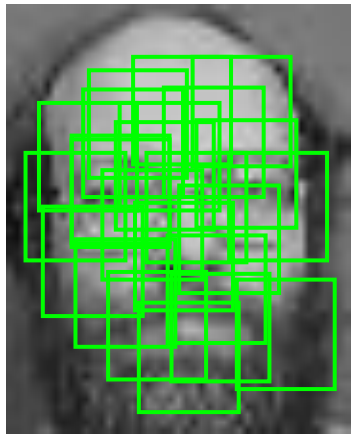


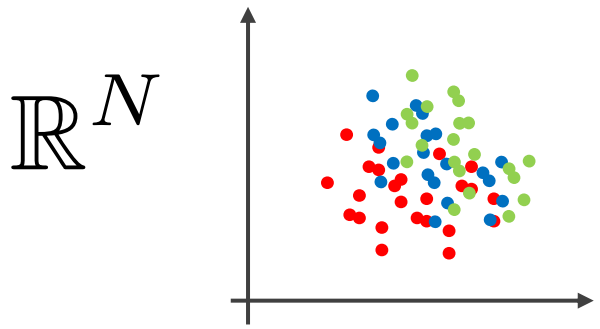


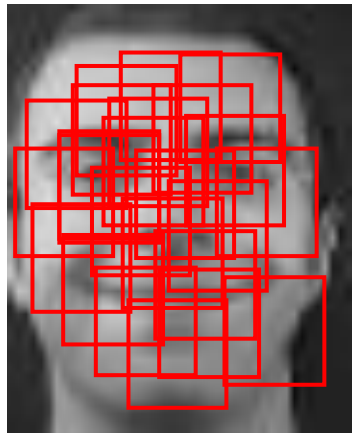

$\mathbb{R}^N$

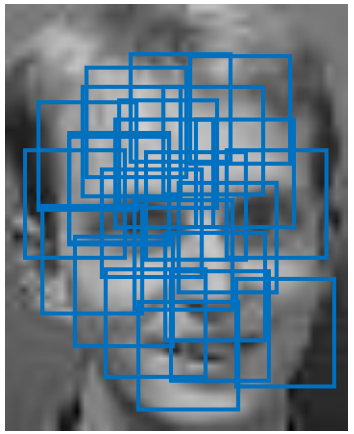


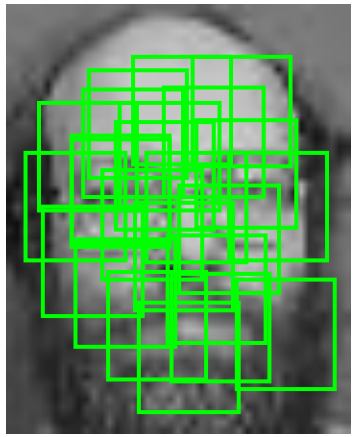





$\mathbb{R}^N$

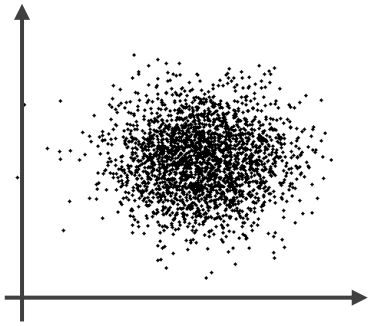






for all images

$\mathbb{R}^N$



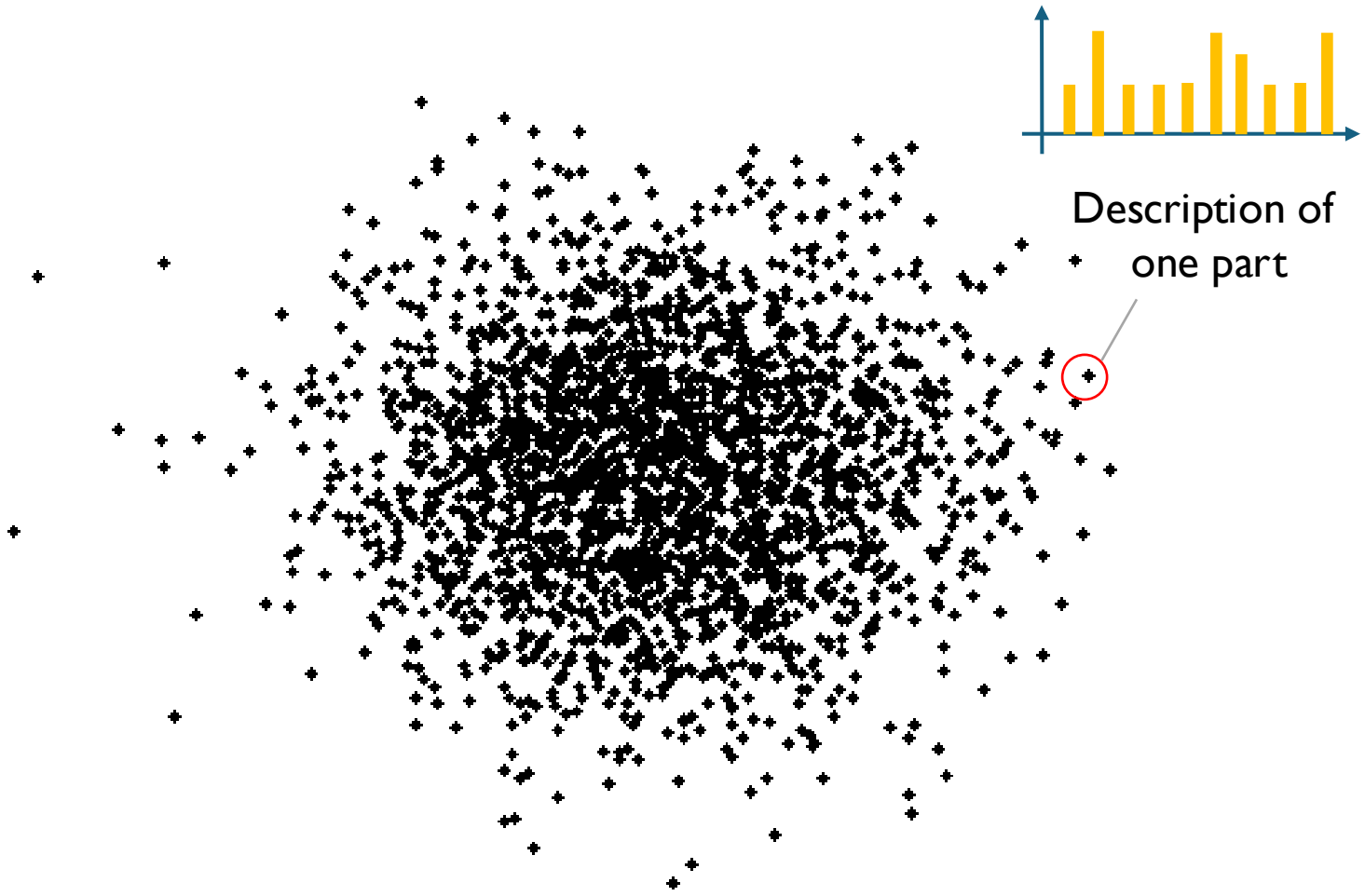
$\mathbb{R}^N$

Description of
all parts

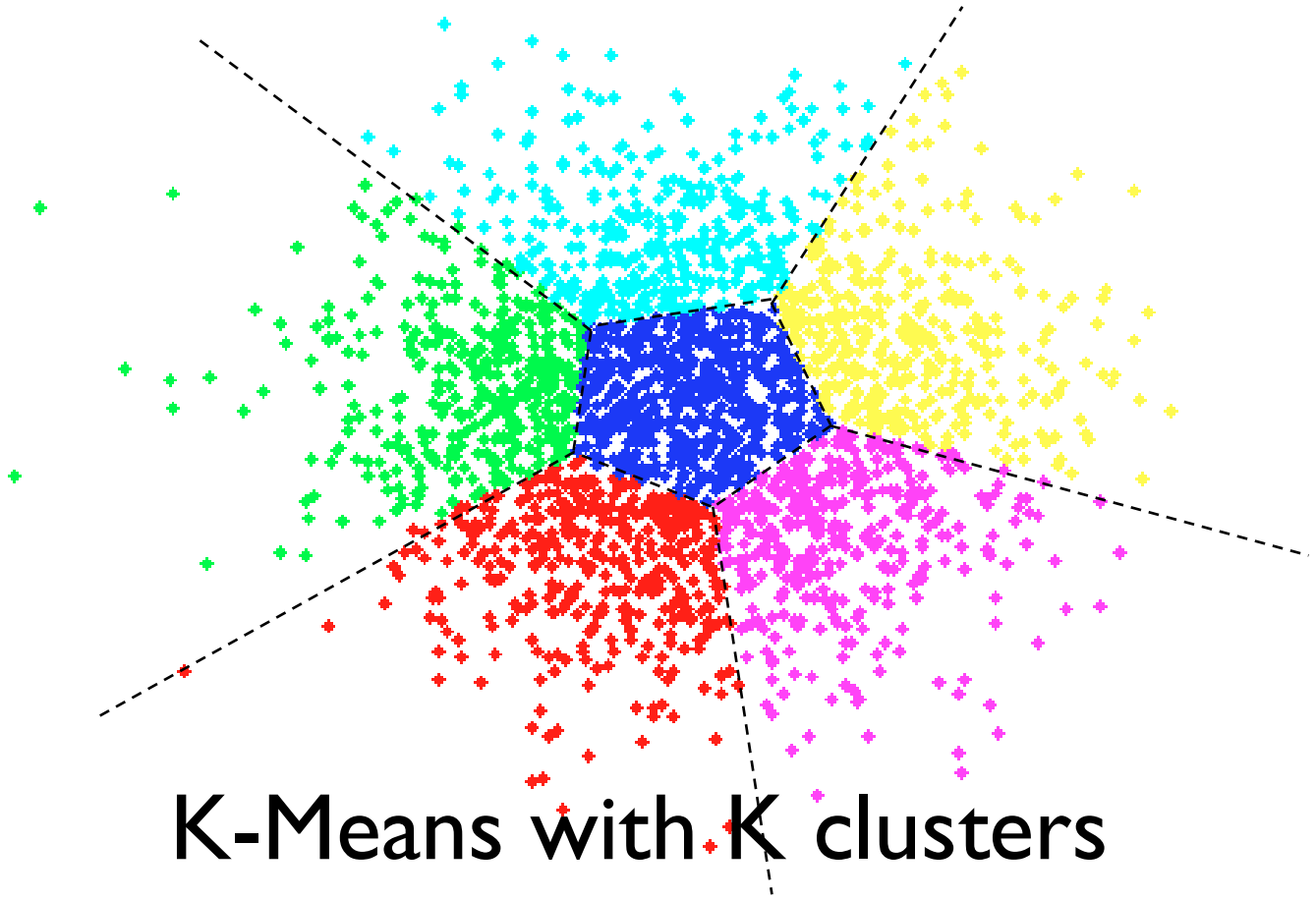


$\mathbb{R}^N$

Description of all parts

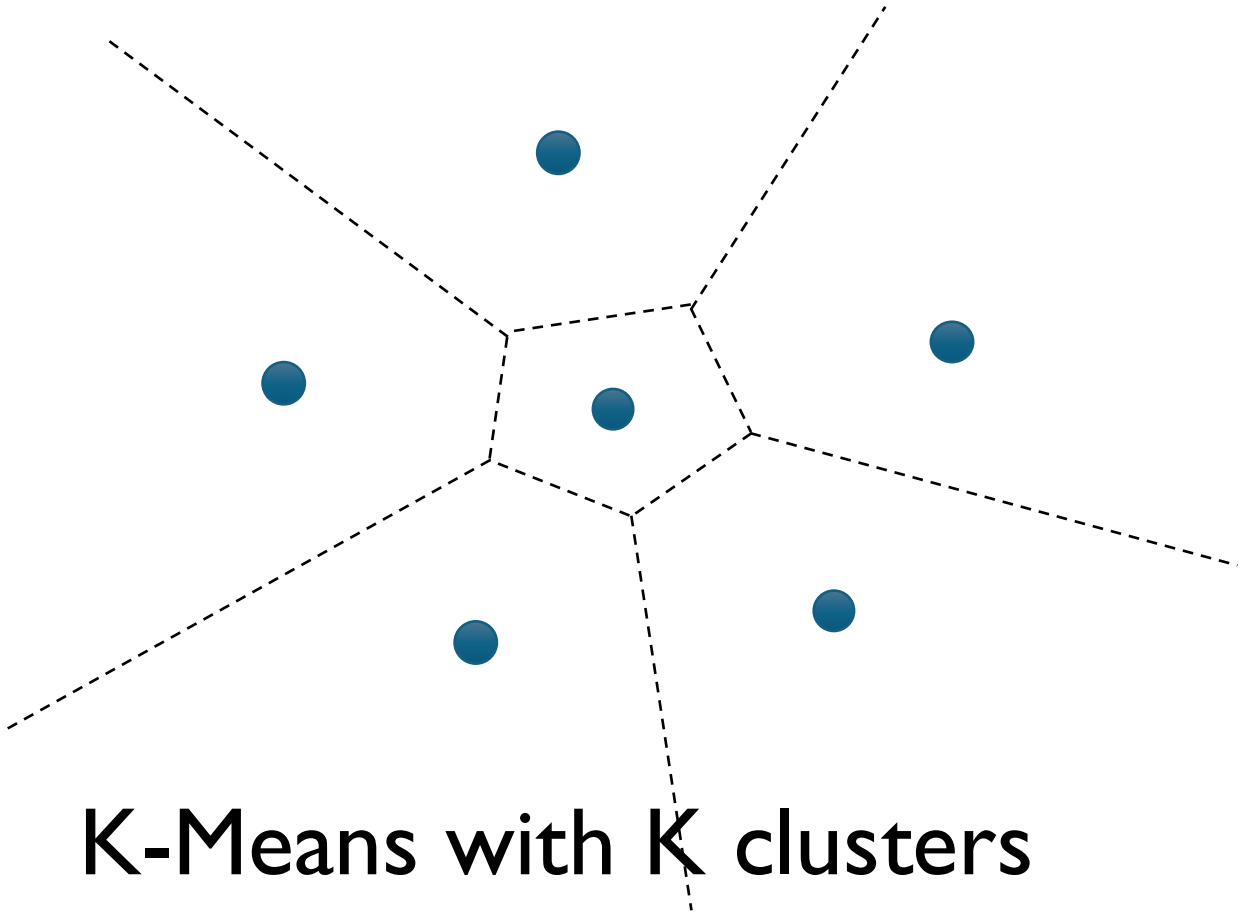


Visual Vocabulary



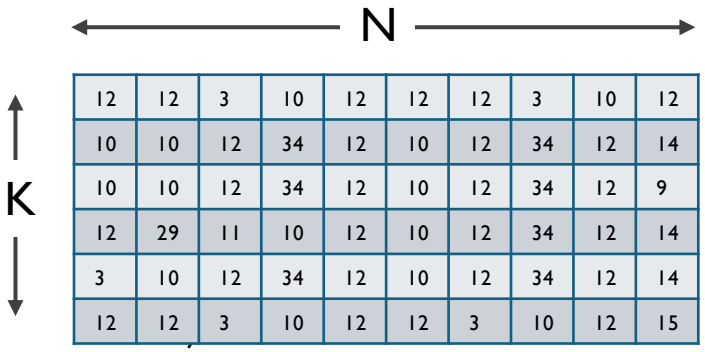
K-Means with K clusters

Visual Vocabulary

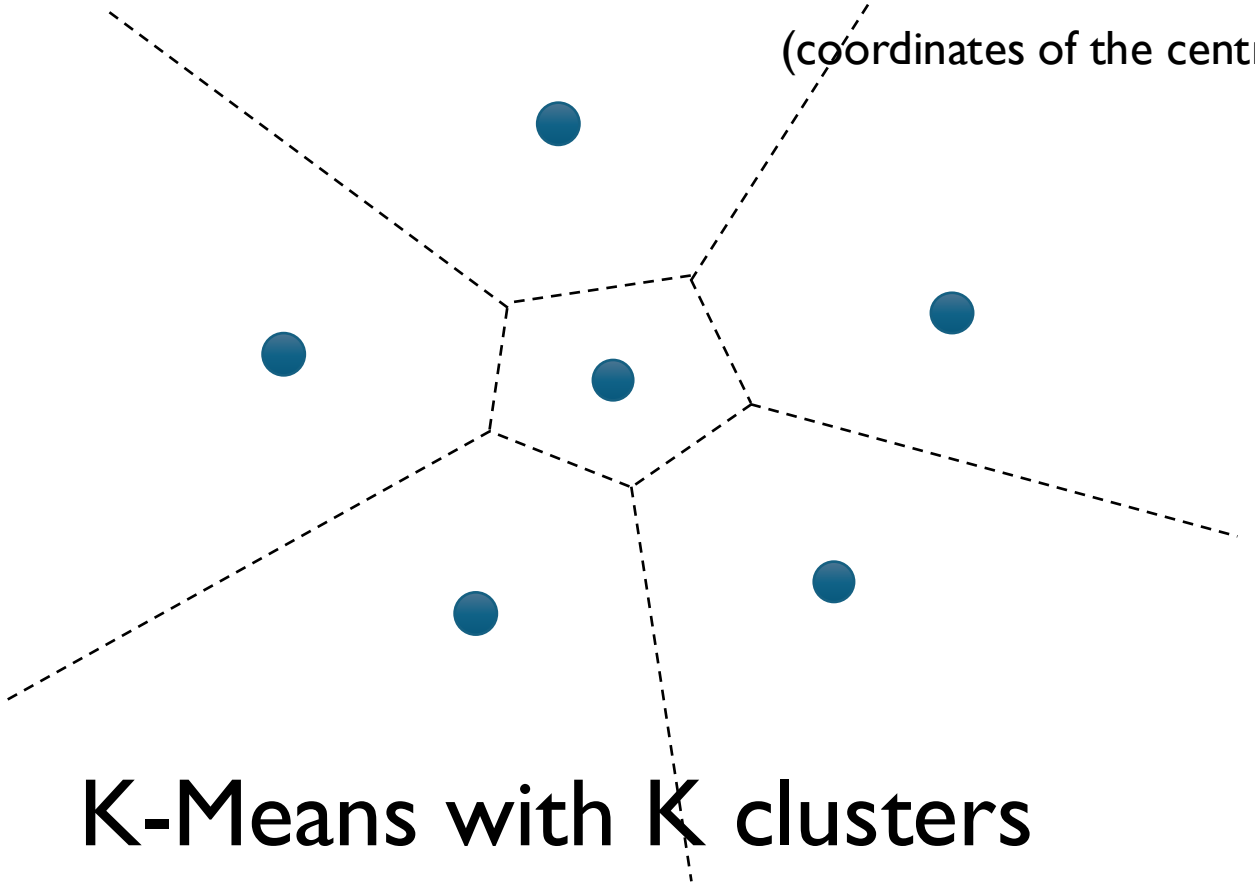


K-Means with K clusters

Visual Vocabulary

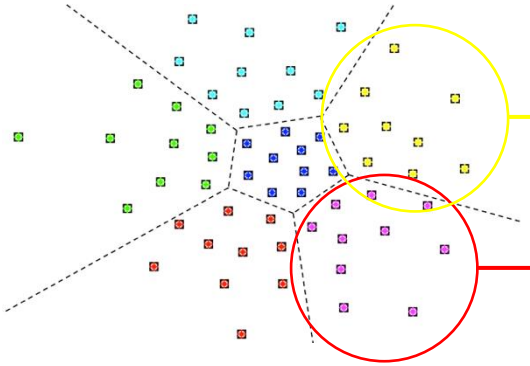


(coordinates of the centroids)



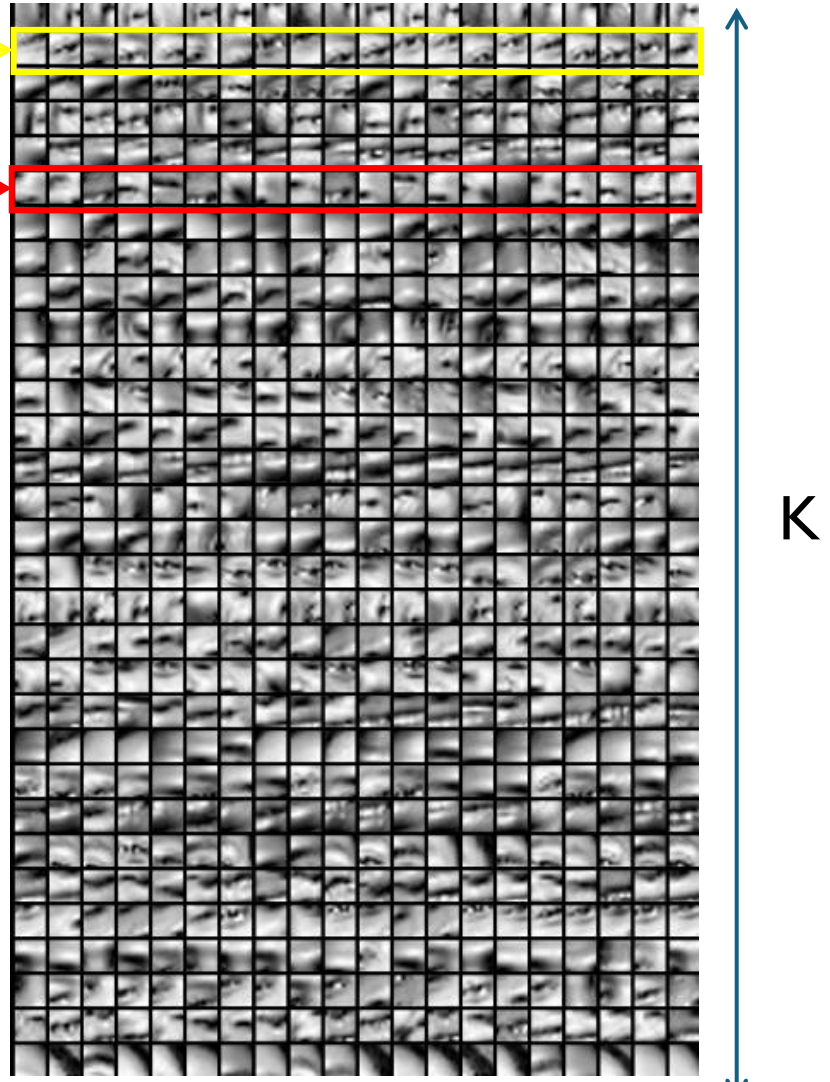
K-Means with K clusters

Example



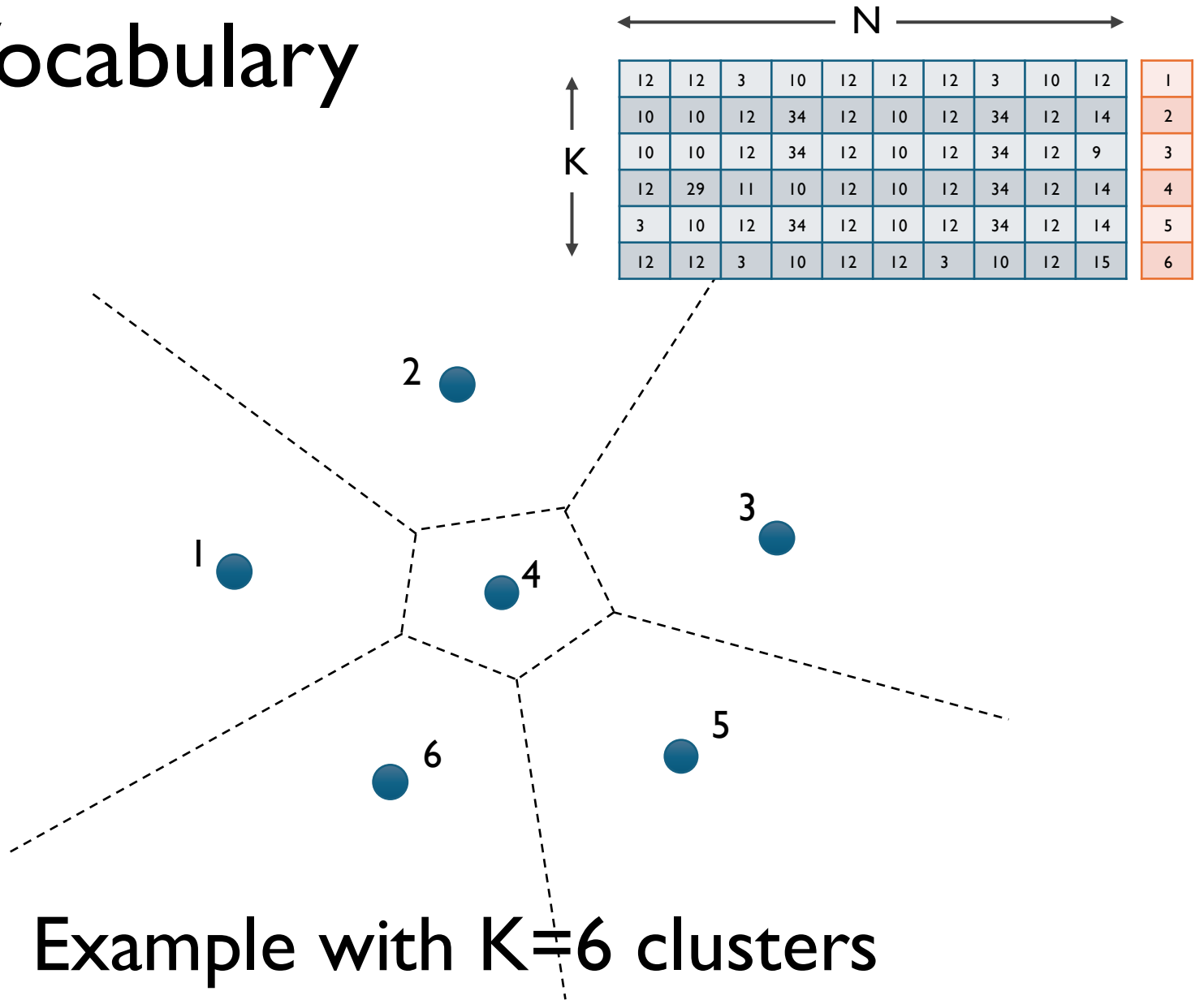
John



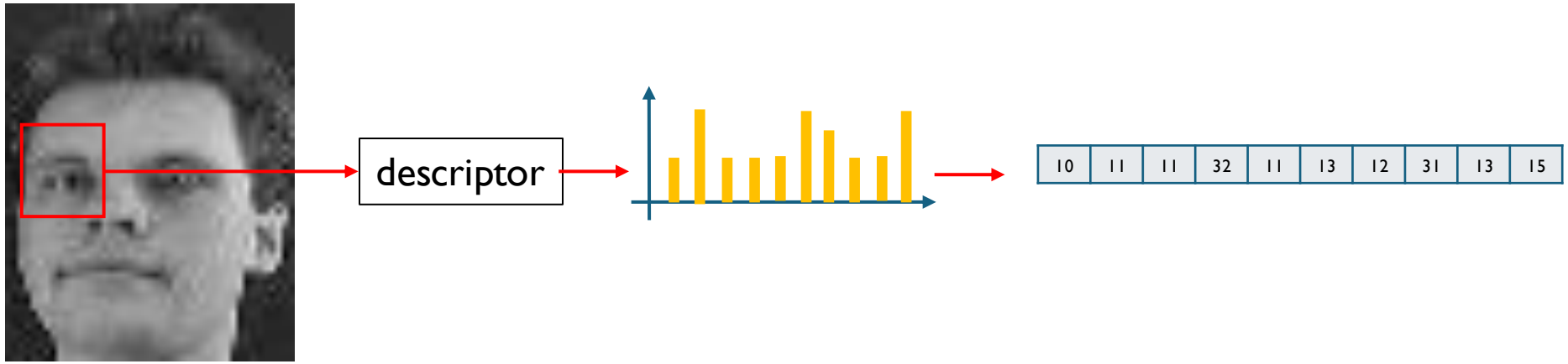


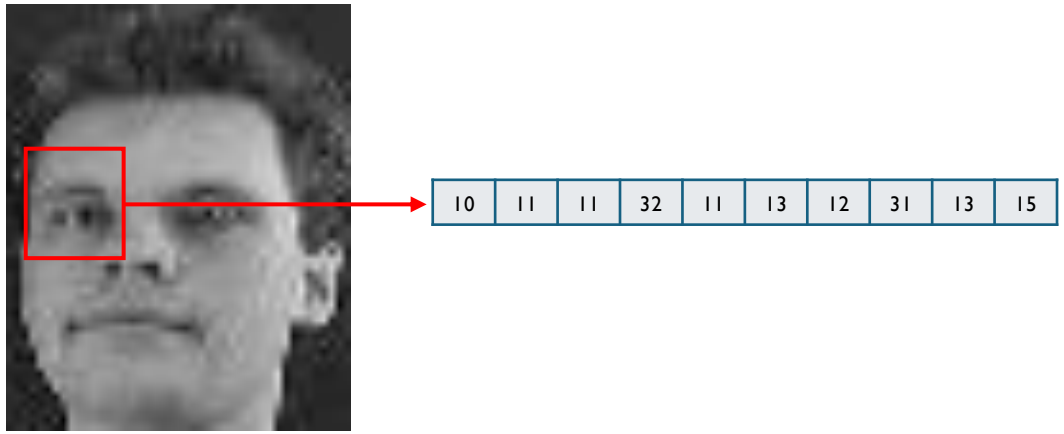
K

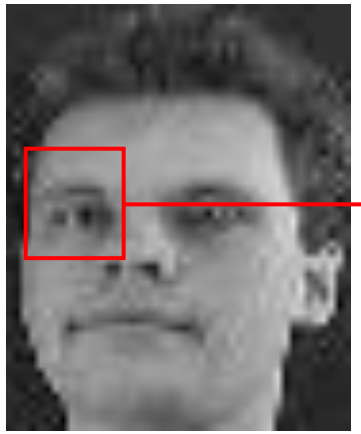
Visual Vocabulary



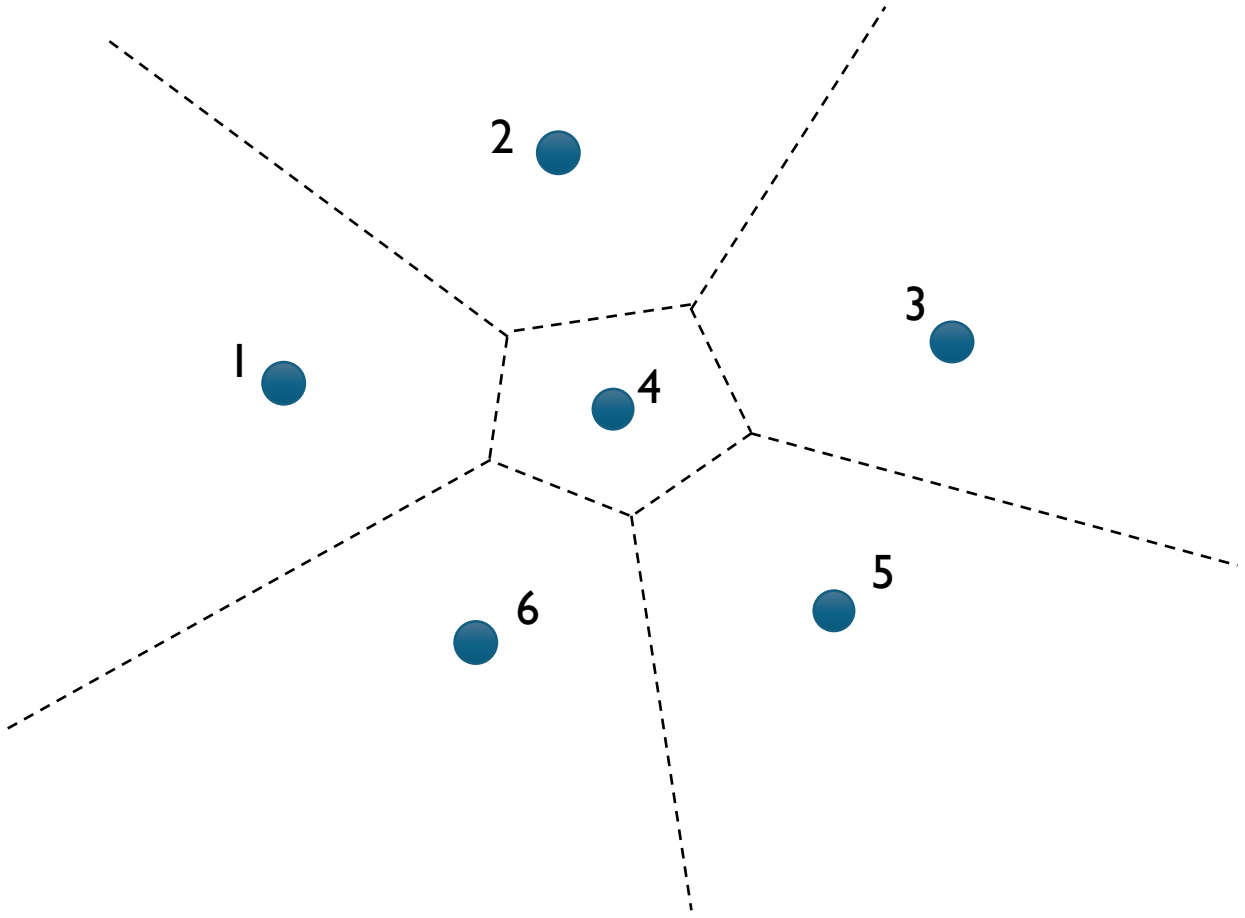
Example with K=6 clusters

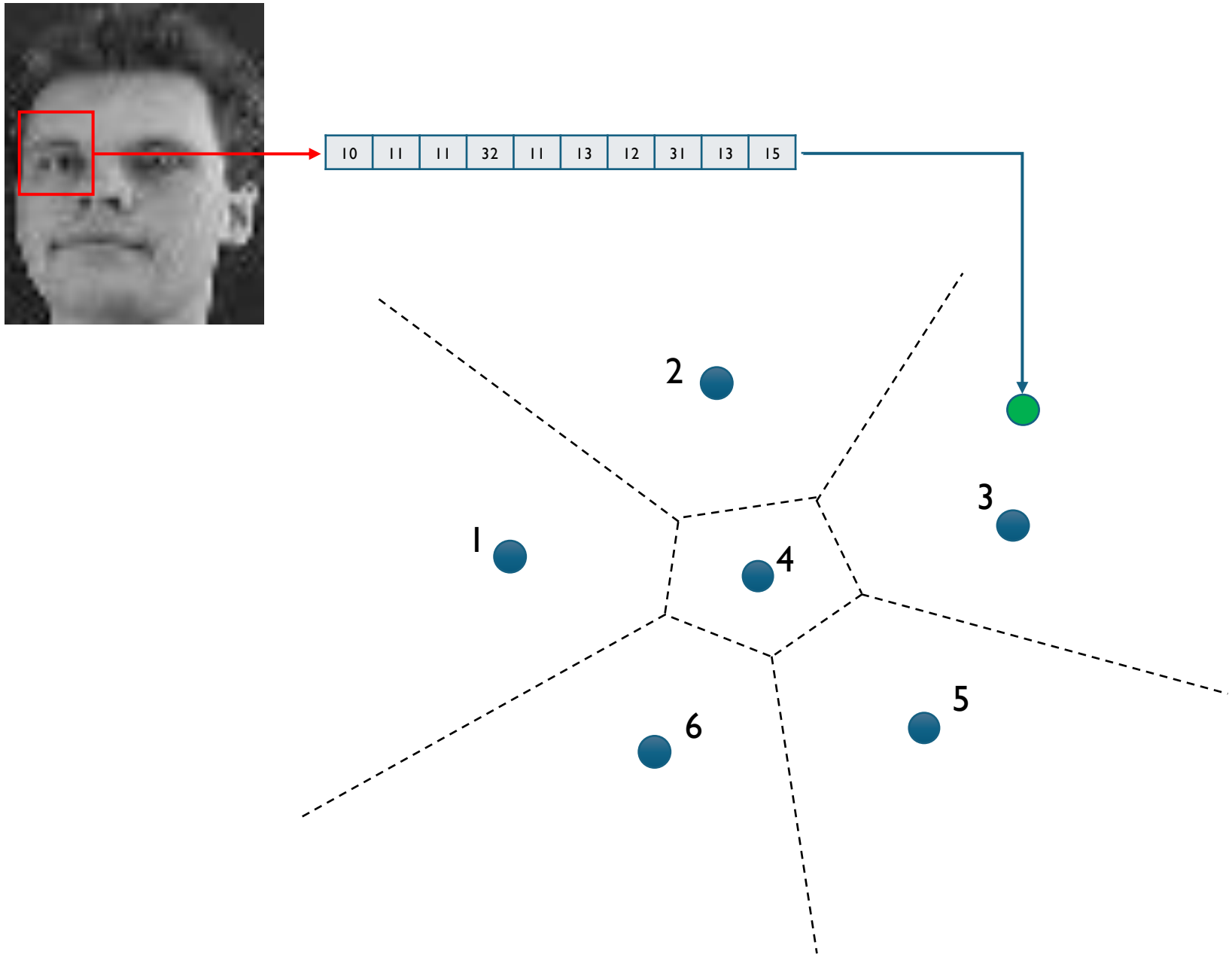

descriptor
10
11
11
32
11
13
12
31
13
15

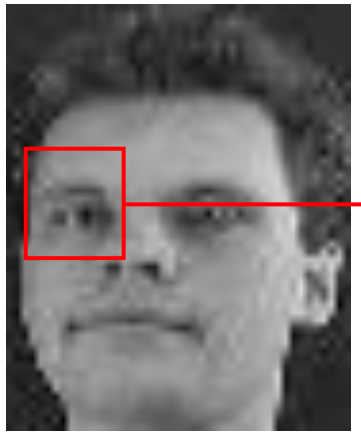

10
11
11
32
11
13
12
31
13
15



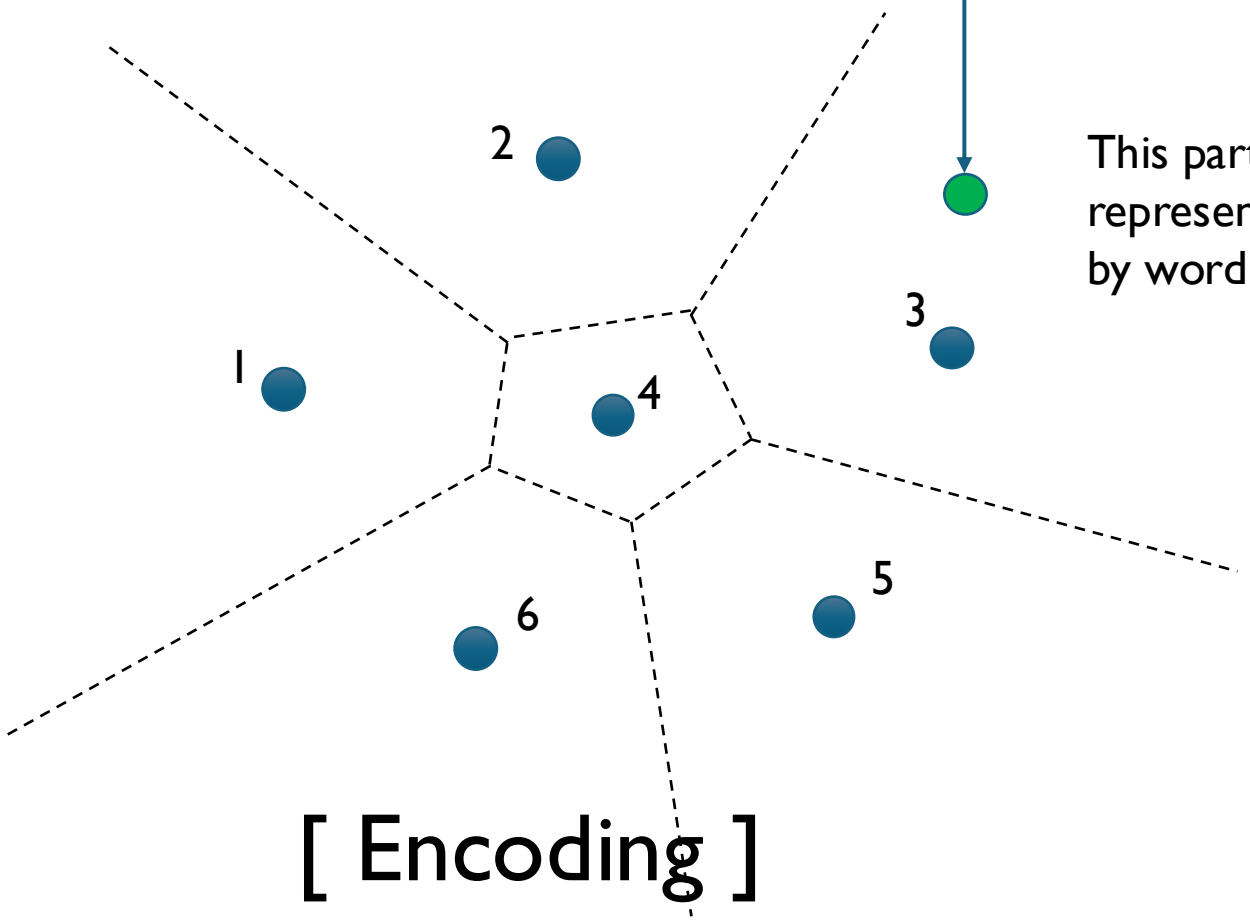
10	11	11	32	11	13	12	31	13	15




10
11
11
32
11
13
12
31
13
15
1
2
3
4
5
6

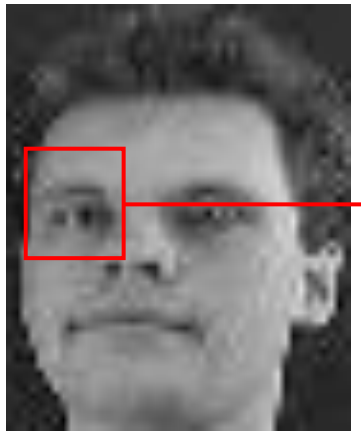






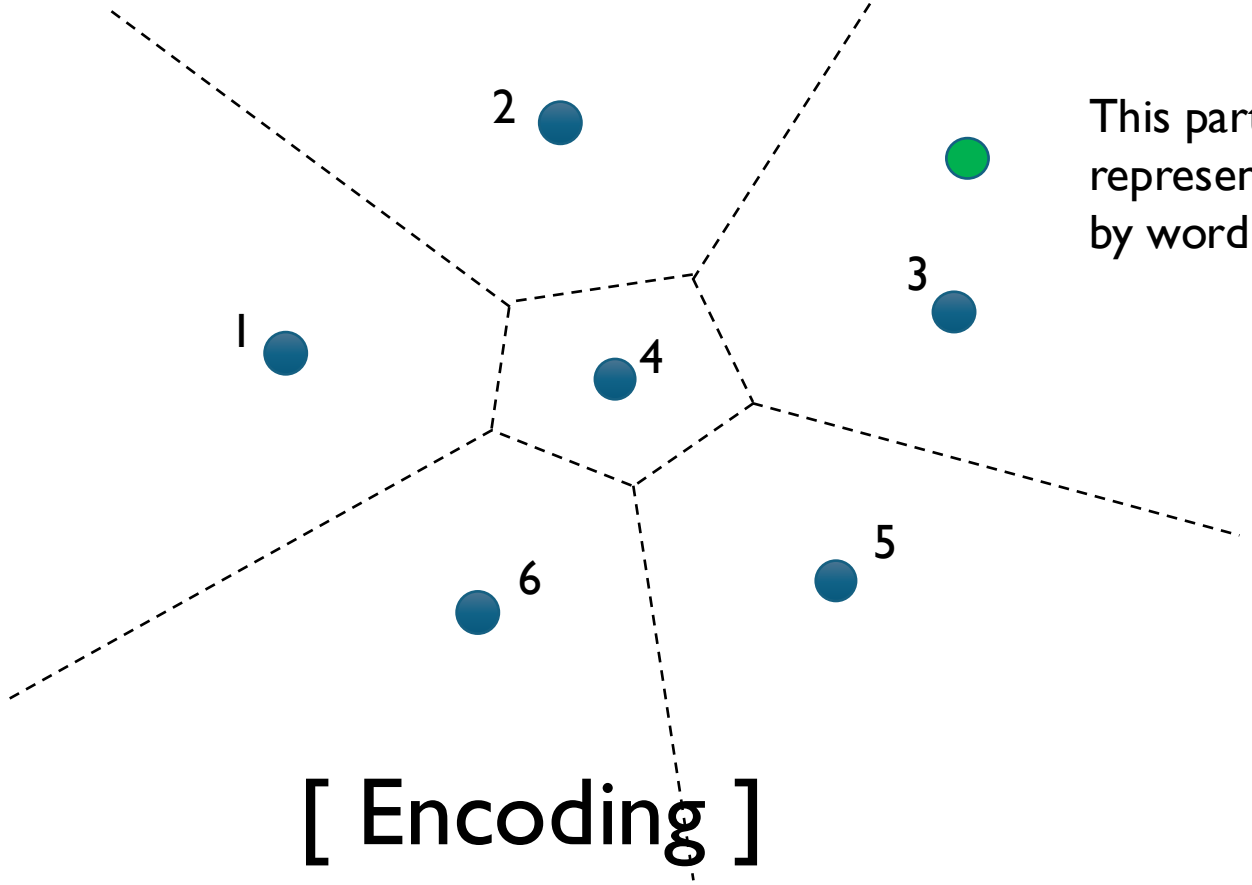
This part is represented by word 3

[Encoding]



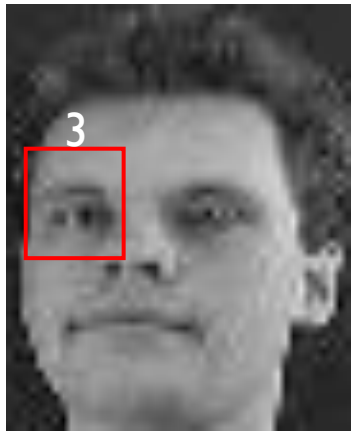
10	11	11	32	11	13	12	31	13	15	

12	12	3	10	12	12	12	3	10	12	1
10	10	12	34	12	10	12	34	12	14	2
10	10	12	34	12	10	12	34	12	9	3
12	29	11	10	12	10	12	34	12	14	4
3	10	12	34	12	10	12	34	12	14	5
12	12	3	10	12	12	3	10	12	15	6

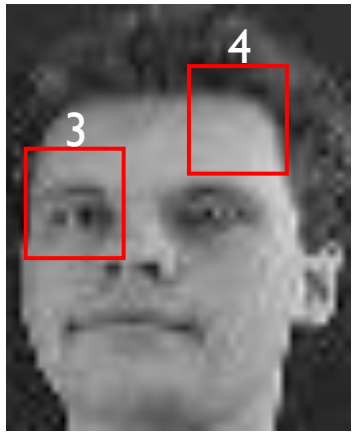


This part is represented by word 3

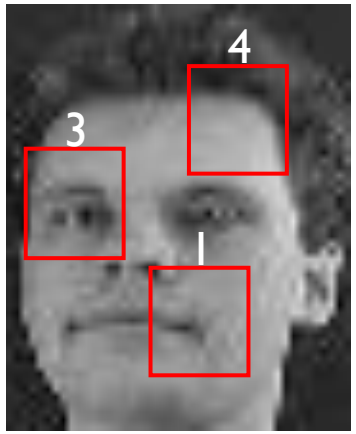
[Encoding]



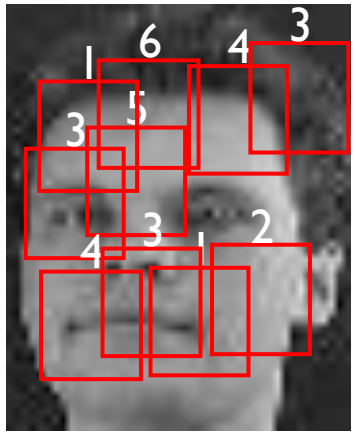
[Encoding]



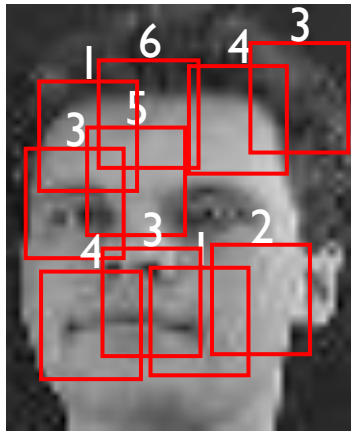
[Encoding]



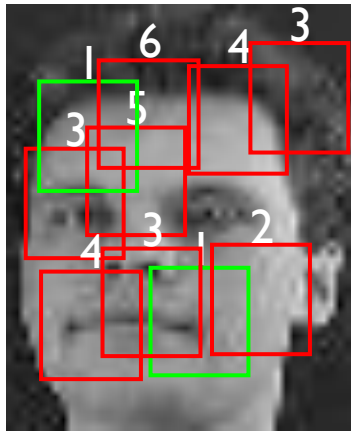
[Encoding]



[Encoding]

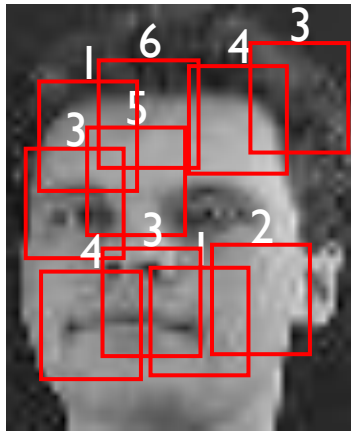


How many words ‘1’ are in this image?

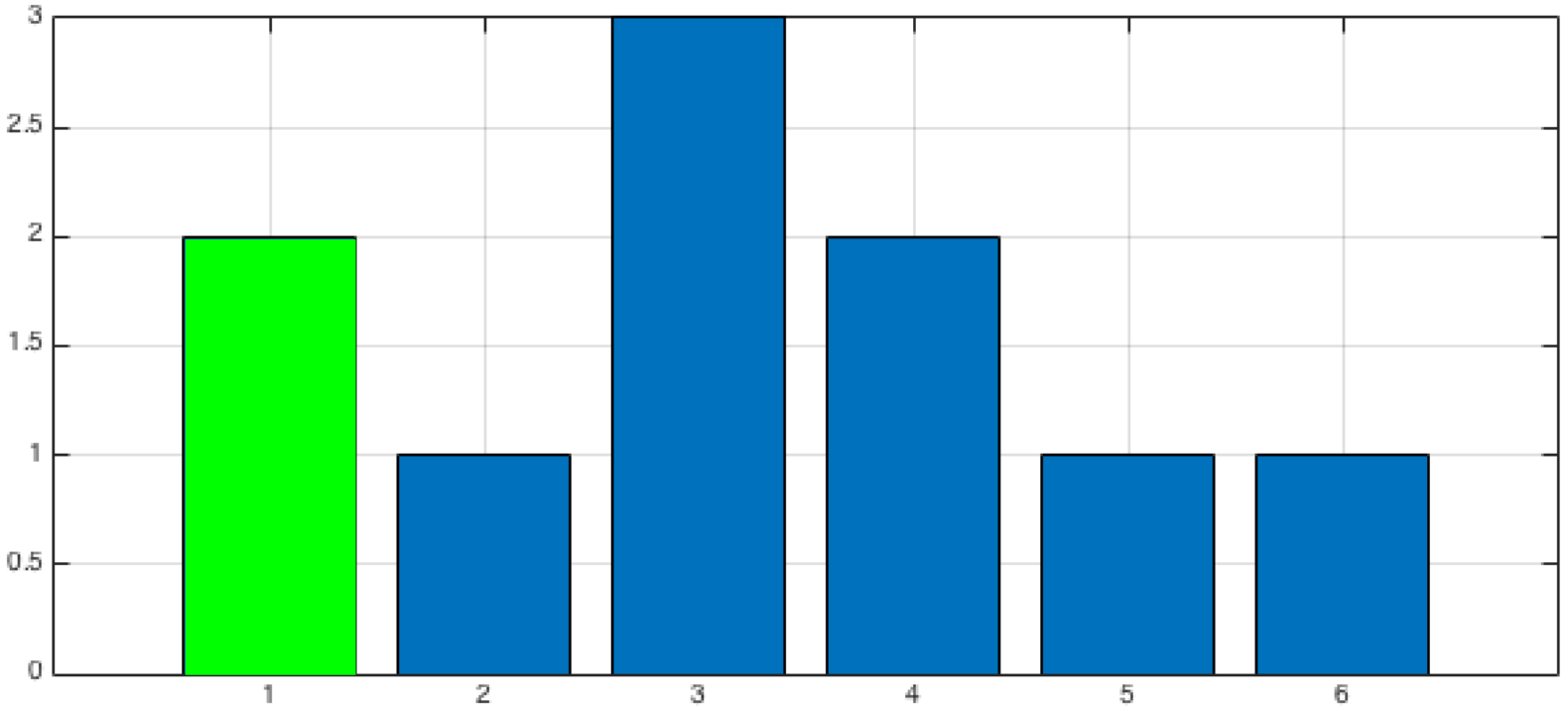


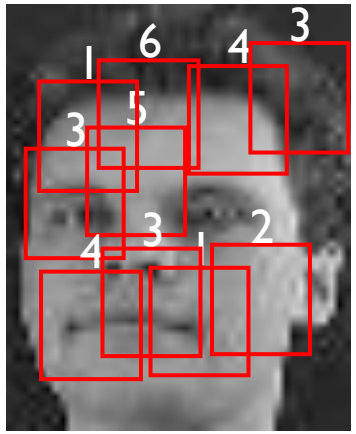
How many words ‘1’ are in this image?

Answer: 2

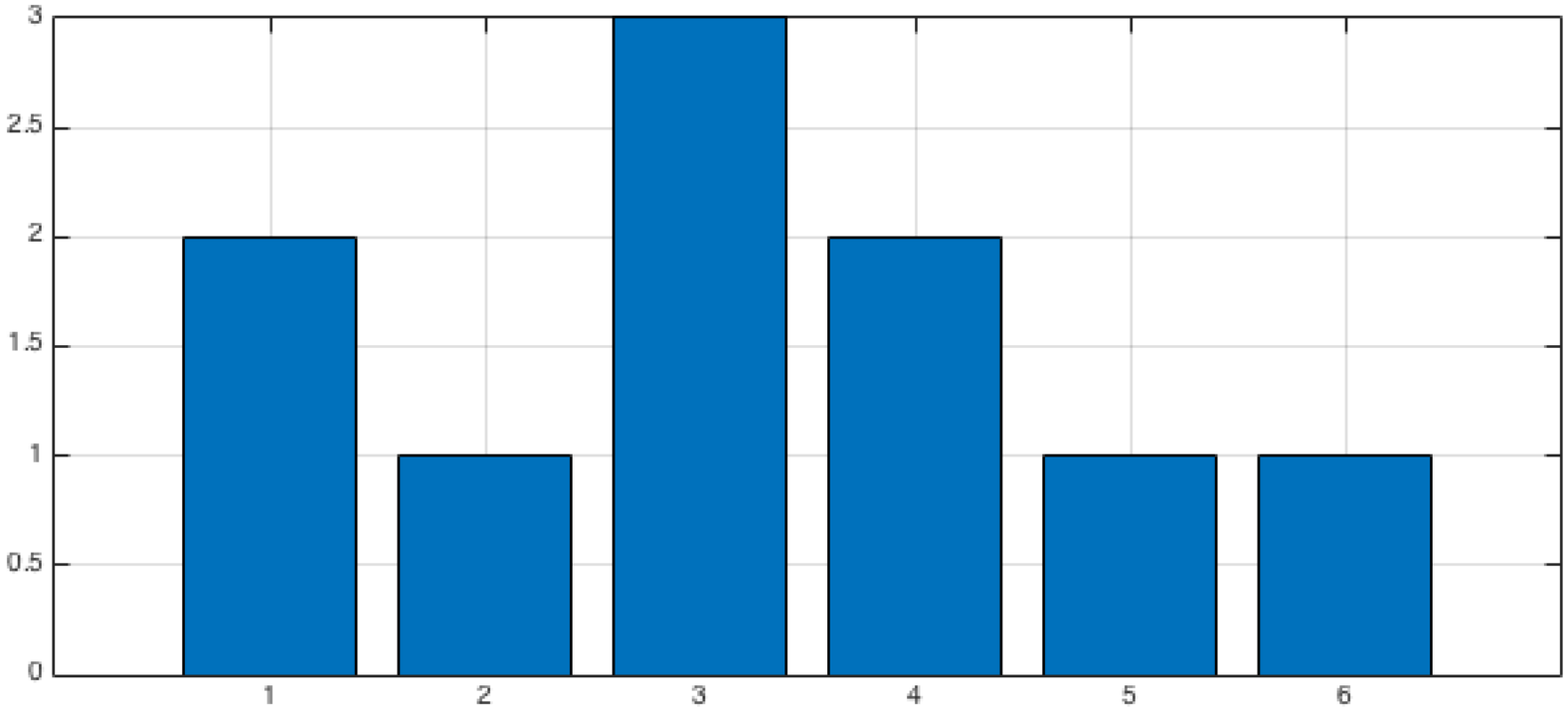


Histogram





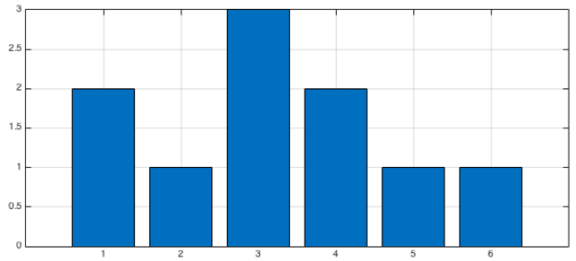
Histogram

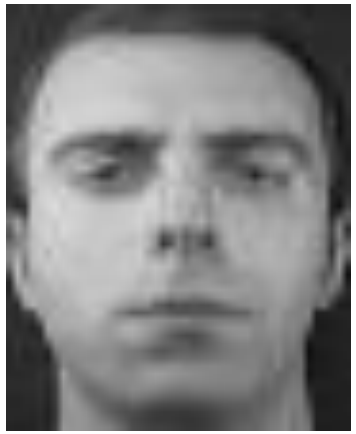




Histogram

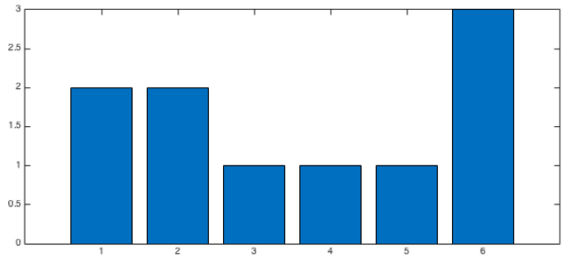






Histogram

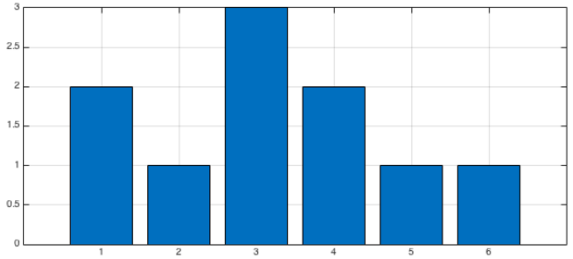


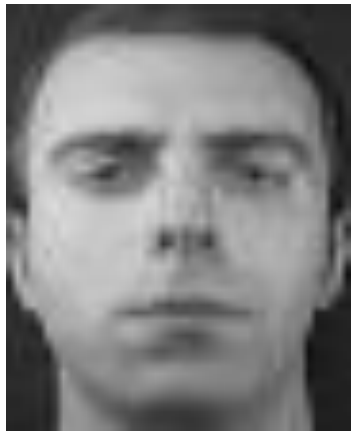


different objects have different histograms

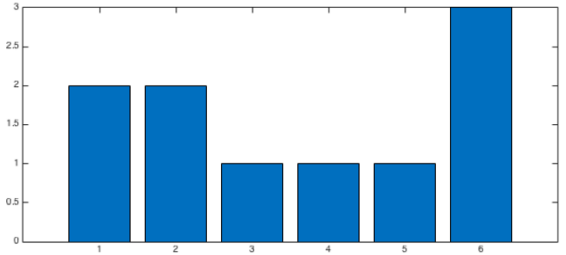


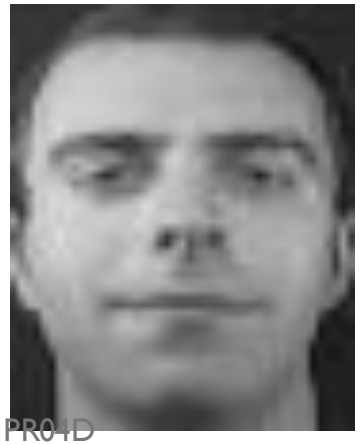
Histogram



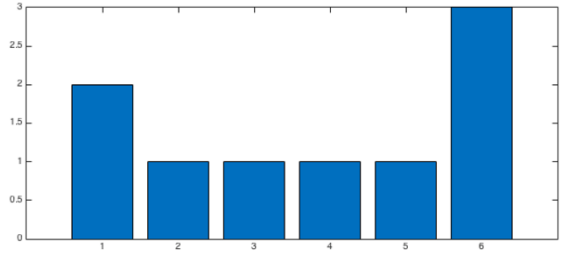


Histogram





Histogram



same objects have similar histograms

Summary

PART 1: VISUAL VOCABULARY

1. Collect representative images
2. Extract descriptors (example LBP, HOG, HoG, etc.)
3. K-means (Visual Vocabulary of K words)

Summary

PART 1: VISUAL VOCABULARY

1. Collect representative images
2. Extract descriptors
3. K-means (Visual Vocabulary of K words)

PART 2: CLASSIFIER DESIGN

1. For each image of the training set
   - Extract descriptors and encode them using the Visual Vocabulary
   - Build the histogram

2. Train a classifier using the new representation (histogram) of each image

3. Repeat for testing set and test the classifier

Ejemplo
Reconocimiento de Tigres
(Biometría)



TRAINING

TRAINING



HoG de cada una de los Tigres

K-means de todos los HoG

$\mathcal{D}$

12	12	3	10	12	12	12	3	10	12
10	10	12	34	12	10	12	34	12	14
10	10	12	34	12	10	12	34	12	9
12	29	11	10	12	10	12	34	12	14
3	10	12	34	12	10	12	34	12	14
12	12	3	10	12	12	3	10	12	15

TRAINING



HoG de cada una de los Tigres

K-means de todos los HoG

$\mathcal{D}$ Coordenadas de K centroides

12	12	3	10	12	12	12	3	10	12
10	10	12	34	12	10	12	34	12	14
10	10	12	34	12	10	12	34	12	9
12	29	11	10	12	10	12	34	12	14
3	10	12	34	12	10	12	34	12	14
12	12	3	10	12	12	3	10	12	15

K

Diccionario de K palabras visuales

N

TRAINING

Para cada tigre del entrenamiento

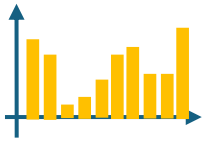


HoG → Codificación → Histograma

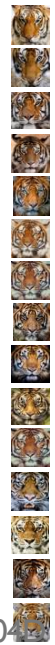
$\mathcal{D}$

K

5	4	1	2	3	4	...	6



$\mathbf{X}$



5	4	1	2	3	4	...	6
6	3	2	3	4	3	...	5
5	2	2	4	3	2	...	4
4	1	4	3	2	1	...	4
3	2	3	2	1	2	...	3
2	3	2	1	2	3	...	2
1	4	1	2	3	4	...	1
2	5	2	3	4	5	...	2
3	6	3	4	5	6	...	3
4	5	4	5	6	5	...	4
5	6	5	6	5	6	...	5
6	7	6	5	6	7	...	6
5	8	5	6	7	8	...	5
6	9	6	7	8	9	...	6
7	0	7	8	9	0	...	7

Histogramas

$\mathbf{y}$

1
1
1
2
2
2
3
3
3
4
4
4
5
5
5

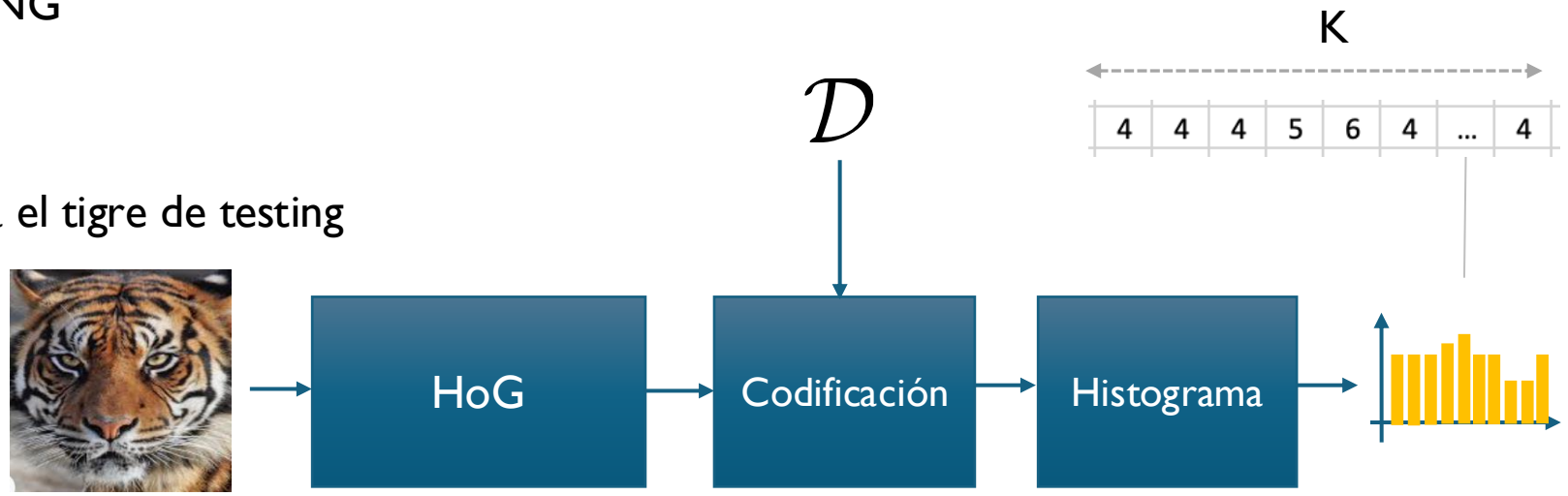
Labels

Entrenar Clasificador → Clasificador Entrenado

TESTING

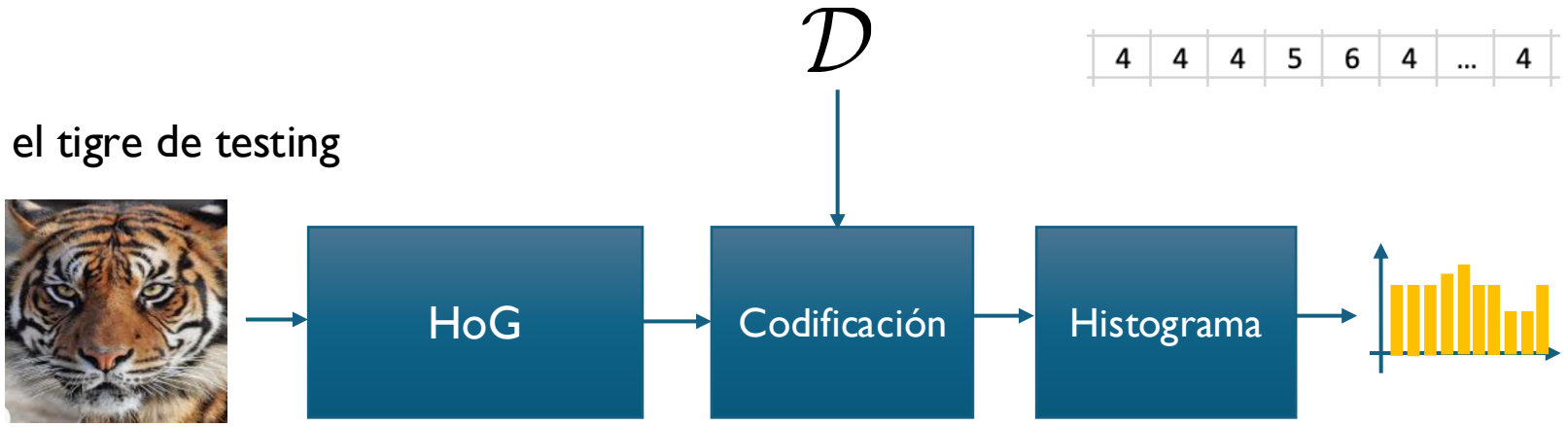
TESTING

Para el tigre de testing



TESTING

Para el tigre de testing





Se puede usar este método para reconocer vacas?